Subsidy Pass-Through and Market Power: Evidence from Production-Based Solar Subsidies

Xiaolin Zhou

October 18, 2026

Abstract

Two production-based solar subsidies in Massachusetts paid residential adopters for the electricity their systems generate, one through tradable certificates sold at market prices and its successor through a fixed rate per kWh set at approval. The programs differed in the form, predictability, and size of the payment. I build a subsidy measure for each roof from its modeled generation, which allows me to absorb the timing of each subsidy with year-month fixed effects and identify pass-through from differences in the subsidy across roofs installed by the same installer in the same town and year. Pass-through is about 85% in both programs. The difference between them is −0.02 (standard error 0.085), which rules out the gap of 0.26 that a comparison of the programs over time suggests. That gap reflects the timing of each subsidy relative to falling installation costs, not the design of the subsidy. Because variation across roofs removes the supply side from the pass-through formula, the shortfall below one bounds installer market power. Demand under SMART is log-concave, and installer markups are at most 2–4% of price at the point estimates and stay at or below about 5% even if a fifth of the variation in modeled generation is error. Unlike the over-shifting found in other settings, high pass-through here points to a nearly competitive market.

1 Introduction

Government subsidies for renewable energy technologies have become a cornerstone of climate policy, and residential solar receives particularly substantial support. In the United States, federal, state, and local governments subsidize residential solar installations through tax credits, rebates, net metering incentives, and other mechanisms (Borenstein, 2017). The federal government alone spent $7.5 billion on solar subsidies in fiscal year 2022, almost all of it through tax credits (U.S. Energy Information Administration, 2023). These subsidies are meant to lower consumers' net cost of adoption and to accelerate the deployment of clean energy technologies, replacing fossil-fuel generation and the negative environmental externalities it causes. Whether they lower costs for consumers depends on how much of the subsidy is passed through to buyers rather than captured by suppliers. Theory predicts that incidence depends on the elasticities of supply and demand and on market structure. Where installers have market power, they can capture part of the subsidy through higher pre-subsidy prices, which reduces adoption and raises the fiscal cost of each unit of new capacity.

This paper studies two production-based programs in Massachusetts that paid residential adopters for the electricity their systems generate. Under the Solar Carve-Out II program (2014–2018), owners earned tradable Solar Renewable Energy Certificates (SRECs) sold at market prices. Its successor, the Solar Massachusetts Renewable Target (SMART) program, pays a fixed rate per kWh that is set at application and declines as capacity blocks fill. The programs differ in the form of the payment, in how predictable it is, and in size: the average Solar Carve-Out subsidy is about 3.4 times the average SMART subsidy per kW ($1,923 versus $566 in present value). Each difference gives a reason to expect pass-through to differ, and a direct comparison seems to confirm it: baseline pass-through is 76% under Solar Carve-Out and 102% under SMART. I show that this comparison is misleading, and I ask what the pass-through rate that remains reveals about installer market power. To my knowledge, this paper is the first to estimate the pass-through of a certificate-based and a

fixed-rate production subsidy and to compare the two within the same market.[1]

The data cover more than 39,000 host-owned residential systems installed between 2014 and 2023. The analysis rests on a subsidy measure built for each roof. Because both programs pay per kWh, a project's subsidy depends on how much electricity it will generate. I merge regulatory data from the Massachusetts Department of Energy Resources (DOER) with Lawrence Berkeley National Laboratory's Tracking the Sun database, geocode each project's street address, and use the National Renewable Energy Laboratory's PVWatts calculator to model each system's annual generation from its location, size, tilt, azimuth, and equipment. Unlike an upfront rebate, whose amount the program computes and records, a production subsidy depends on generation the data do not report, so I model generation for each roof and compute the present value of its subsidy. Prior studies identify pass-through mainly from changes in rebate levels over time and across utilities (Pless and van Benthem, 2019). I identify it from differences in the subsidy across roofs at the same point in time. This variation is large: about 91% of the variance in modeled capacity factors is within towns, so a town-level measure would discard most of the variation in subsidy value across roofs.

The subsidy varies for two reasons. First, it changes over time, with SREC prices under Solar Carve-Out and with the step-downs in SMART rates. Second, expected generation differs across roofs, a source of variation that exists in both programs and does not depend on the policy calendar. I regress the price buyers pay net of the subsidy on the subsidy, with granular fixed effects that control for system quality, local market conditions, and time trends. In the preferred specification, the timing of the subsidy is absorbed, and pass-through is identified only from differences in the subsidy across roofs installed by the same installer in the same town and year.

The first finding is that the two designs pass through the subsidy at the same rate. When

[1]Earlier comparisons in other settings find that the form of an incentive can affect pass-through: car buyers capture 70 to 90 percent of a customer rebate but 30 to 40 percent of a dealer discount (Busse, Silva-Risso and Zettelmeyer, 2006), and the point at which a diesel tax is collected changes its pass-through (Kopczuk et al., 2016).

only differences in generation across roofs identify pass-through, the estimate is 0.86 under Solar Carve-Out and 0.85 under SMART. The difference is −0.02 with a standard error of 0.085, so differences larger than about 0.18 in either direction are ruled out, including the baseline gap of 0.26. That gap reflects the timing of the two programs relative to falling installation costs. Year-month fixed effects alone remove most of it, and the SMART estimates that rely on timing exceed one, which log-concave demand cannot produce unless marginal cost falls with volume. Buyers capture about 85 cents of each dollar of subsidy, whether it is paid through a certificate market or through a fixed rate, and whether it covers 45% or 15% of the price.

The second finding concerns market power. I use the sufficient-statistics framework of Weyl and Fabinger (2013) as Pless and van Benthem (2019) do, but in reverse. Pless and van Benthem find more-than-complete pass-through together with convex demand and conclude that solar markets in California are imperfectly competitive. The variation across roofs that delivers the first finding is well suited to the reverse test. A difference in the subsidy across buyers of the same installer does not move the installer along its supply curve, so the shortfall of pass-through below one reflects pricing power and the curvature of demand, or cost differences across roofs, but not rising marginal cost. Following their analysis, I estimate demand at the ZIP-code level and find that demand under SMART is log-concave, so market power would push pass-through below one. With pass-through of 85%, installer markups under SMART are at most 2–4% of price at the point estimates, and they stay at or below about 5% even if a fifth of the variation in modeled generation is error. Buyers receive about 80 to 85 percent of the surplus the subsidy creates. Reduced-form checks agree: pass-through does not fall when installers are busy and does not vary significantly with local installer concentration.

This paper contributes to three literatures. The first estimates the pass-through of solar subsidies. Most of this evidence comes from upfront rebates, especially the California Solar Initiative, and finds that buyers capture most but not all of the subsidy: 78% for host-owned

systems in California (Pless and van Benthem, 2019), 84% in Connecticut (Gillingham and Tsvetanov, 2019), and close to complete pass-through in other work on California (Dong, Wiser and Rai, 2018). Pass-through also depends on ownership and market structure: Pless and van Benthem (2019) find more-than-complete pass-through for third-party-owned systems. For the federal investment tax credit, reported prices also reflect the incentive to overstate system costs (Podolefsky, 2013). I provide estimates for production-based subsidies paid through certificates and through fixed rates and compare the two mechanisms within one market. Once the timing of the subsidy is absorbed, they pass through the same share, and the apparent difference between them comes from the timing of each subsidy relative to costs.

The second literature studies tax and subsidy incidence more broadly. Estimates range from incomplete to more than complete, and pass-through tends to rise with competition: from 13% in the least competitive Medicare Advantage markets to 74% in the most competitive (Cabral, Geruso and Mahoney, 2018), and, for fuel taxes on Greek islands, from 0.4 in monopoly markets to 1 in markets with four or more competitors (Genakos and Pagliero, 2022). Other studies find full pass-through of hybrid vehicle tax credits (Sallee, 2011), 70 to 90 percent of manufacturer cash rebates (Busse, Silva-Risso and Zettelmeyer, 2006), half to three-quarters of the E85 subsidy (Lade and Bushnell, 2019), more-than-complete pass-through of alcohol taxes (Kenkel, 2005), and high but incomplete pass-through of carbon costs (Fabra and Reguant, 2014).[2] Pass-through has been used to measure incidence under imperfect competition (Ganapati, Shapiro and Walker, 2020) and to detect market power (Pless and van Benthem, 2019). I use it the other way around: with log-concave demand, high pass-through bounds market power, and variation in the subsidy across roofs, which

[2]Vehicle subsidies have received particular attention, including hybrid vehicle subsidies with endogenous quality and costly bargaining (Gulati, McAusland and Sallee, 2017), California's electric vehicle subsidies (Muehlegger, Rapson et al., 2018), and electric vehicle subsidies across countries (Barwick et al., 2023). Related work studies the incidence of agricultural subsidies on farmland rents (Kirwan, 2009) and the welfare effect of a consumer subsidy restricted by price ceilings (Fan and Zhang, 2022). Incomplete pass-through can also reflect local costs and markup adjustment (Nakamura and Zerom, 2010), and firms that set prices nationally may respond differently to local cost shocks (Butters, Sacks and Seo, 2022).

removes the supply side, gives the bound directly.

The third literature shows that subsidy design affects how buyers respond. Solar adoption responds strongly to rebates (Hughes and Podolefsky, 2015; Crago and Chernyakhovskiy, 2017), buyers discount production-based payments heavily (De Groote and Verboven, 2019), and upfront subsidies induce more adoption while production subsidies are more efficient (Burr, 2016).[3] My results add that, once the timing of the subsidy is accounted for, a certificate-based subsidy and a fixed-rate subsidy are passed through at about the same rate.

Section 2 describes the two programs, Section 3 the data and the construction of the subsidy, and Section 4 the empirical strategy. Section 5 compares pass-through across the two programs, Section 6 estimates demand and derives the bound on market power and the division of surplus, and Section 7 concludes. The appendix describes the data construction, reports the baseline specifications, further robustness checks, a comparison of pooled and separate estimation, and the rule-based designs, and discusses other explanations for the baseline gap and price discrimination across roofs.

2 Institutional Background

When a household in Massachusetts installs a solar system, one of two production-based subsidies is available, depending on the policy regime at the time of installation. If a qualifying system entered commercial operation between 2008 and November 2018, the owner was eligible to receive certificates under the Solar Carve-Out program. If it entered commercial operation after November 2018, the owner is entitled to an incentive payment through the electricity bill, on top of net metering, under the SMART program. This section describes the two programs, and Section 3 describes the data and the method I use to compute the

[3]Other studies compare the effectiveness of residential solar incentives (Matisoff and Johnson, 2017), estimate how residential solar demand depends on net metering and electricity prices (Ros and Sai, 2023), and characterize optimal subsidies for residential solar (Colas and Reynier, 2026). Consumer subsidies can also lower production costs through induced innovation (Gerarden, 2023), and the phase-out of a subsidy shapes the dynamics of demand (Hu, Yin and Zhao, 2023).

subsidy.

2.1 Solar Carve-Out

The Solar Carve-Out is part of Massachusetts's Renewable Portfolio Standard (RPS) and requires that a certain percentage of electricity come from solar resources. The program launched on January 1, 2010, and its requirement increased over time until it reached a capacity target of 400 MW. The state expanded the program in April 2014 with Solar Carve-Out II, which set a total goal of 1,600 MW of solar PV capacity.

Energy suppliers, including utilities and competitive energy suppliers, are required to purchase Solar Renewable Energy Certificates (SRECs) to demonstrate compliance or make an alternative compliance payment (ACP). The NEPOOL Generation Information System (NEPOOL GIS) tracks solar generation and issues SRECs quarterly to solar generation unit owners at a rate of 0.45–1.0 SREC per MWh, depending on system characteristics (the SREC factor). For example, a system smaller than 25 kW authorized to interconnect before January 8, 2017, receives a factor of 1.0. Each qualified solar generation unit is entitled to receive SRECs for 40 calendar quarters as long as the unit remains operational. After 40 calendar quarters, the system earns Renewable Energy Certificates (RECs) instead, which are much less valuable than SRECs.[4]

Solar generation unit owners can either market and trade SRECs independently or join an aggregator that harvests and trades SRECs on their behalf. Trading platforms and aggregators typically charge fees of 1–10% of the proceeds depending on system size, with small residential systems (under 50 kW) commonly paying 10%. The Massachusetts Department of Energy Resources (DOER) holds an annual fixed-price clearinghouse auction from May 16 to June 15 for SRECs not sold during the previous year. The auction does not guarantee

[4]The vintage of a certificate is the calendar year in which the electricity is generated. For instance, electricity generated during 2014 earns MA2014(II) certificates, where "MA" denotes Massachusetts, "2014" the generation year, and "II" the program phase, Solar Carve-Out II. Certificates of different vintages form distinct market segments and can trade at different prices, depending on factors such as program caps, ACP levels, and the balance between supply and demand within each vintage year.

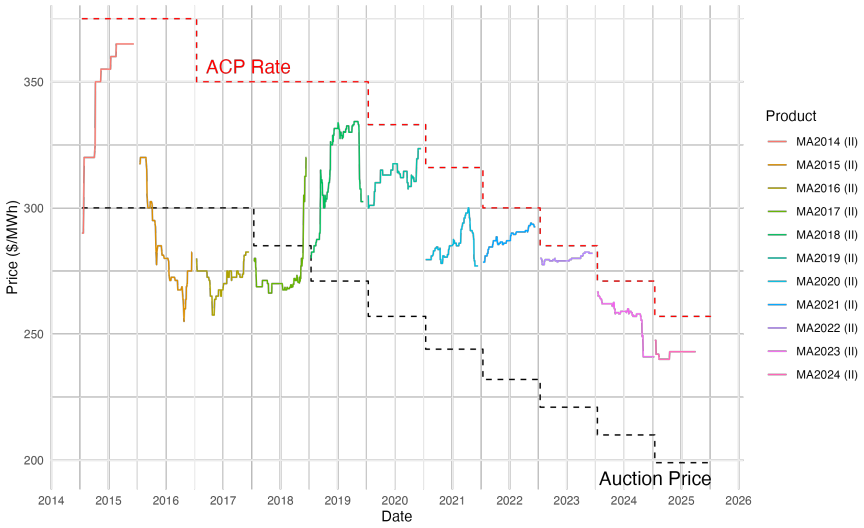


Figure 1: SREC II Market Prices with ACP and Auction Prices. The auction price is plotted gross of the 5% auction fee. An owner selling through the auction receives 95% of the plotted value ($285/MWh in 2014–2016). Each vintage is shown over its trading window, from July 15 of the vintage year to July 14 of the following year.

a sale: certificates that remain unsold through it can still be traded in the market for the next two compliance years. The ACP thus caps SREC prices, but the auction price is not a floor. In the Solar Carve-Out's launch year, the ACP was set at $600/MWh, while the net clearinghouse auction price was $285/MWh (a $300 price less the 5% auction fee).[5] For Solar Carve-Out II, the ACP and auction price schedules were published at launch, whereas under the original program the ACP was adjusted over time. Publishing the schedules narrowed the range of likely SREC prices but did not remove price risk: as Figure 1 shows, market prices sat below the auction price for several years.

Figure 1 shows historical market prices for SREC II certificates alongside the ACP and clearinghouse auction prices. In the first few years, market prices were consistently below auction prices because of oversupply. In later years, market prices for SRECs followed the declining trend of the ACP. The net clearinghouse price remained constant at $285/MWh from 2014 to 2016 and then fell by 5% per year.[6]

[5]Under the original program, market prices for SRECs tracked closer to the ACP, and returns exceeded twice the retail electricity price in 2010.

[6]The net clearinghouse price stood at $189.05/MWh in 2024.

2.2 SMART

DOER created the Solar Massachusetts Renewable Target (SMART) program to further encourage solar development in Massachusetts after the Solar Carve-Out II target was met. The program launched on November 26, 2018. Under SMART, the three investor-owned utilities (IOUs) in Massachusetts (Eversource, National Grid, and Unitil) compensate solar system owners for the electricity their systems generate, in exchange for the RECs attached to this generation.

The actual incentive rate for a given system is determined by four factors: the base compensation rate, household income status, whether the system includes a storage component, and the value of energy. The base compensation rate starts high and declines on a predetermined schedule as capacity blocks fill, and both block sizes and rate schedules vary by utility. The value of energy is set from three-year average retail electricity prices (including delivery and transmission charges) at the time of application. The SMART incentive rate equals the base compensation rate minus the value of energy. Small residential solar generation unit owners receive payments through the utility equal to this rate multiplied by their monthly generation, in addition to the bill savings their behind-the-meter systems already provide, and they can cash out a negative balance monthly. As electricity prices rise and capacity blocks advance, the SMART incentive rate decreases until it reaches zero.[7]

Table 1 shows the base compensation rates for generation units of 25 kW AC or less, by utility.[8]

SMART was most favorable early in its life cycle, when the incentive rate was considerably higher than the price of RECs. In my sample, the SMART incentive rate is at most \$0.20/kWh, with a median of about \$0.066/kWh, while RECs traded at around \$0.04/kWh.[9]

[7] System owners who do not wish to enroll in SMART can instead participate directly in the REC market, and DOER advises them to do so when the SMART rate becomes low. I do not observe how many owners opt out because they are in the regulatory data if they do so.

[8] There are eight blocks for the three large utilities but only two and four for Nantucket and Fitchburg, respectively. Fitchburg is Unitil's operating company, and Mass Electric and Nantucket Electric are National Grid's operating companies in Massachusetts.

[9] Source: SRECTrade trading platform.

Table 1: SMART Base Compensation Rates for Generation Units of 25 kW AC or Less

Utility	Low income	Block 1	2	3	4	5	6	7	8
Fitchburg	Y	\$0.358	\$0.326	\$0.298	\$0.272	—	—	—	—
	X	\$0.311	\$0.284	\$0.259	\$0.236	—	—	—	—
Mass Electric	Y	\$0.358	\$0.344	\$0.330	\$0.317	\$0.304	\$0.292	\$0.280	\$0.269
	X	\$0.311	\$0.299	\$0.287	\$0.275	\$0.264	\$0.254	\$0.244	\$0.234
Nantucket	Y	\$0.391	\$0.328	—	—	—	—	—	—
	X	\$0.340	\$0.286	—	—	—	—	—	—
Eversource E.	Y	\$0.391	\$0.375	\$0.360	\$0.346	\$0.332	\$0.319	\$0.306	\$0.294
	X	\$0.340	\$0.326	\$0.313	\$0.301	\$0.289	\$0.277	\$0.266	\$0.255
Eversource W.	Y	\$0.329	\$0.315	\$0.303	\$0.291	\$0.279	\$0.268	\$0.257	\$0.247
	X	\$0.286	\$0.274	\$0.263	\$0.253	\$0.243	\$0.233	\$0.224	\$0.215

Base compensation rates in dollars per kWh by utility and capacity block.

SMART also offers more predictable cash flows than the Solar Carve-Out program, because the rate is fixed and does not fluctuate with market conditions. The subsidy can therefore be computed precisely from modeled generation. Another key difference between SMART and Solar Carve-Out is that SMART requires system interconnection to one of the three IOUs. As a result, households served by municipal utilities are ineligible for the program.

2.3 Comparison of the Two Programs

Table 2 summarizes the two programs. Four differences could affect how subsidy value is divided between buyers and installers. First, the form of payment: an owner under Solar Carve-Out sells certificates whose price is set in a market capped by the ACP, whereas a SMART owner receives a fixed per-kWh rate. Second, the timing of information: the SMART rate is fixed when the application is approved, so the buyer and installer know the subsidy per kWh for ten years, while the SREC price the owner will eventually receive is unknown at the time of purchase. Third, the size of the subsidy: the mean present value is about \$1,923 per kW under Solar Carve-Out and \$566 per kW under SMART, roughly 45% and 15% of the average gross price, with almost no overlap between the two distributions. Fourth, coverage: Solar Carve-Out is open to all utility territories, while SMART is limited

to customers of the three IOUs.

Table 2: Comparison of the Solar Carve-Out II and SMART Programs

	Solar Carve-Out II	SMART
Period in sample	Jan. 2014–Nov. 2018	Mar. 2019–Sep. 2023
Form of payment	Tradable SRECs	Fixed rate per kWh
Rate	Market price, capped by the ACP	Fixed at approval
Payment term	40 calendar quarters	10 years
Payer	Retail electricity suppliers	The three IOUs
Eligibility	All utility territories	IOU customers only
Mean subsidy (NPV, 8%)	$1,923 per kW	$566 per kW
Share of gross price	~45%	~15%

Subsidy is the present value per kW of DC capacity at an 8% discount rate with 0.5% annual degradation, from the analysis samples. Shares are ratios of the sample mean subsidy to the sample mean gross price per kW (systems of 25 kW or less).

These differences suggest that pass-through could differ between the programs. A larger subsidy that arrives through a volatile market price is harder to price into a quote and may be more likely to strain installers' capacity, which would lower pass-through. A small, certain subsidy is easy to incorporate into a price and is less likely to move installer capacity, which would raise pass-through. Because the programs also differ in years, technology costs, and eligible households, a comparison of the programs over time cannot test these predictions. Section 5.3 tests whether pass-through differs once the timing of the subsidy is absorbed, and Appendix C examines these channels for the baseline gap.

3 Data

3.1 Data Sources and Sample

I obtain the solar installation data from DOER. For SMART projects, the DOER data report compensation rates as well as project characteristics, including location, utility, size, installation cost, installer, and technical details. The technical details, which include az-

imuth, tilt, and inverter rating, are essential for modeling generation accurately. The compensation rate reported in the data is the actual SMART incentive rate that system owners receive, which reflects any adders and the value of energy.

For Solar Carve-Out II projects, DOER reports similar variables but without technical details. To model annual generation as accurately as for SMART, I merge the DOER data with the Tracking the Sun database published by Lawrence Berkeley National Laboratory (LBNL).[10] All projects in my data appear in Tracking the Sun, along with their technical details. I match projects across the two sources on operation date, size, and total cost, which gives an almost one-to-one match (Appendix A). SREC prices are not available in the project data. I construct an SREC price series from a major SREC trading platform, SRECTrade, and from the price support regulations published on DOER's website.

The SREC sample spans January 1, 2014, to November 26, 2018, the period when Solar Carve-Out II was open to applications. The SMART sample begins on March 25, 2019, marking the debut of the incentive program, and ends on September 20, 2023, by which time most projects received zero compensation and the number of applications had dropped sharply. For SMART projects, I date each project by its Commercial Operation Date (Appendix A).

The sample includes only host-owned, small-scale (≤25 kW DC) residential projects without battery storage, for three reasons. First, the general net metering program has a capacity cap, so projects installed after the cap was reached might not be eligible for net metering. Small systems are exempt from the cap, so restricting the sample to them ensures that every system is on net metering. Second, only host-owned projects have directly observed prices. For third-party-owned projects, the price reported to DOER is reportedly a "market" value rather than the true cost to system owners (Pless and van Benthem, 2019). Third, installations with battery storage qualify for a storage adder, which raises the incentive, and the storage components would distort the measured cost of the solar system. I also remove

[10]Tracking the Sun is a cleaned dataset that covers more than 60% of solar projects in the United States. LBNL cleans and consolidates information on solar projects from state energy agencies annually. DOER and the Massachusetts Clean Energy Center are both major data providers for LBNL.

observations with a per-kW price above the 99th percentile or below the 1st percentile to exclude outliers.

3.2 Modeling Annual Generation

The empirical analysis in Section 4 regresses the price each buyer pays net of the subsidy on the subsidy itself, so it requires the value of the subsidy for each project. Because both the SMART and Solar Carve-Out II programs are production-based subsidies, the total subsidy depends on how much electricity a system will generate and when the payments are made. SMART payments are made monthly through the utility, so the subsidy is the incentive rate times monthly generation. Under the SREC program, payment comes from selling certificates, which are issued every three months, so the subsidy is the market price of certificates times the number of certificates produced each quarter. For simplicity, I calculate the subsidy on an annual basis, which requires annual generation for each project.

I model annual generation using the PVWatts calculator, a tool developed by the National Renewable Energy Laboratory (now the National Laboratory of the Rockies) for modeling the energy production of grid-connected photovoltaic systems. The calculator combines solar irradiance and meteorological data for a project's location with a set of system parameters to compute monthly generation.

The most important inputs for the calculator are a project's size and location, which determine the area of solar panels and the available solar resource. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained each project's street address from DOER and converted it to coordinates (Appendix A). Figure 2 shows the location of each project in the Solar Carve-Out (panel a) and SMART (panel b) samples. The main difference between the two samples is coverage: households in municipal utility territories are ineligible for SMART. In both panels, the density of projects closely resembles the population distribution of Massachusetts.

Besides size and location, other required parameters include the DC-to-AC ratio, module

type, array type, azimuth, and tilt.[11]

The calculator output includes monthly and annual generation for each project. I summarize it with the capacity factor: a system's annual generation as a share of what it would produce if it ran at full rated capacity all year. Most projects have modeled capacity factors of 12.5–15%, and a few are much lower.[12]

3.3 Validating Modeled Generation

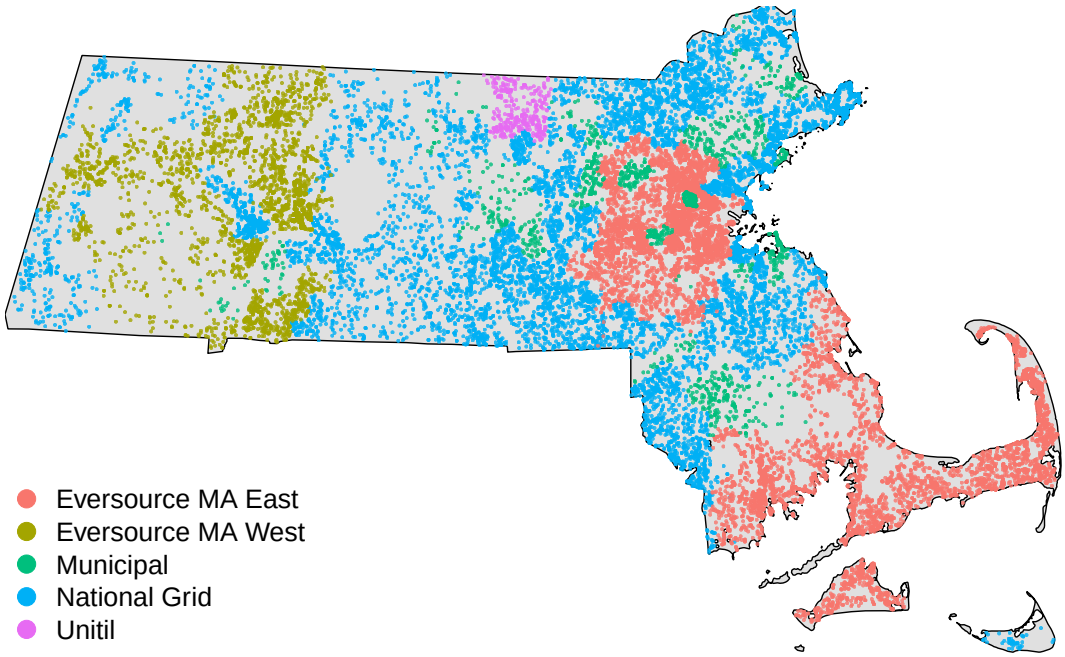
The subsidy is proportional to modeled generation, so its accuracy affects both the regressor and the dependent variable. PVWatts output is a model prediction, not a measurement. I fix the system loss parameter at the calculator default of 14.08%, assume a fixed roof-mount array with standard modules, and use observed size, tilt, azimuth, and DC-to-AC ratio where available. The calculator does not model site-specific losses from shading, snow cover, soiling, and unavailability beyond the default values.

The average modeled capacity factor is 14.3%, above the 13.1% DOER reports from metered generation for the whole Massachusetts fleet in 2010–2023.[13] The level of modeled generation matters less for the pass-through estimates than this gap might suggest. If PVWatts overstated every project's generation by the same factor k, the measured subsidy would be k times the true one. Because the dependent variable subtracts the measured subsidy, the estimated pass-through would then be $1-(1-\rho)/k$, which is closer to one than the true ρ. Even if the whole gap of $k \approx 14.3/13.1 \approx 1.09$ applied to my sample, a true

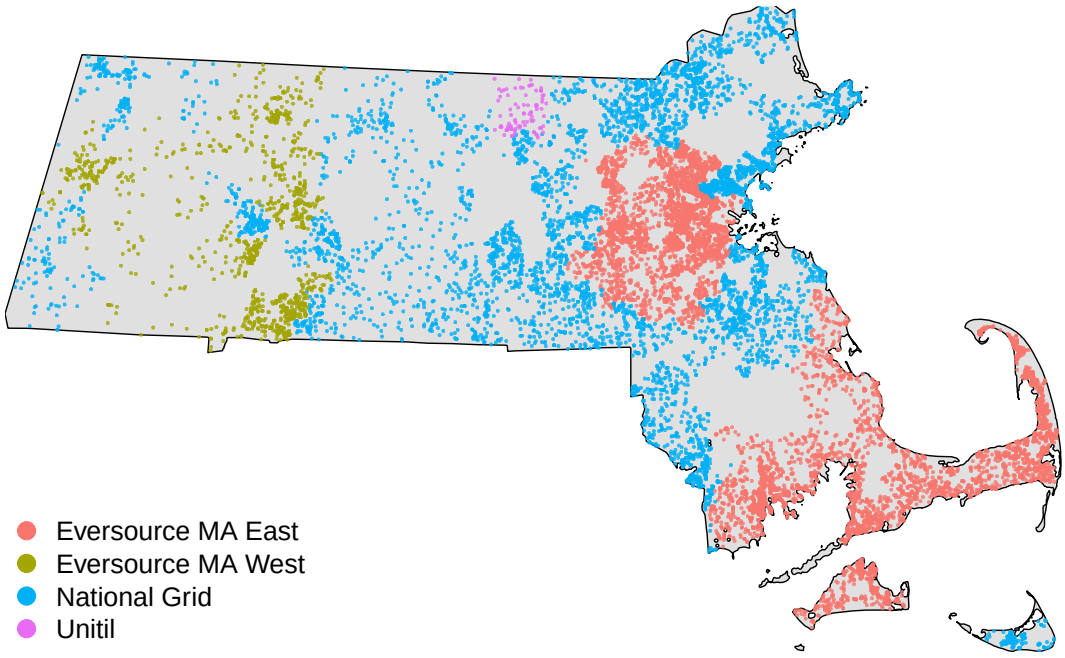
[11] I enter the coordinates as the location input. System size is available for every project, but other parameters are missing for some projects. For a missing parameter, I calculate an implied value. For example, I compute the DC-to-AC ratio for SMART projects from their DC and AC ratings. When no implied value is available, I use the calculator's default values, which represent typical rooftop systems. Fewer than 100 projects require default values.

[12] By entering the technical specifications into the calculator manually, I verified that the low values are correct. They come from suboptimal configurations: a north-facing system, for example, generates much less than a south-facing one, all else equal.

[13] Panel degradation explains only part of this gap. Field evidence suggests that panels lose about 0.5% of output per year (Deline et al., 2024), and applying that rate over the age distribution of the fleet closes roughly a fifth of the difference. The larger part of the gap most plausibly reflects losses that PVWatts does not model. The default snow loss of zero is unrealistic for Massachusetts, and the default shading and soiling losses of 3% and 2% are likely too low given the long pollen season and restrictions on tree removal. The statewide fleet also includes lower-quality installations that predate my sample.



(a) Solar Carve-Out II



(b) SMART

Figure 2: Project Locations by Utility Territory. Each dot is one project in the analysis sample. Municipal utility customers are ineligible for SMART.

pass-through rate of 0.84 would appear as 0.85. Part of the gap comes from older panels in the statewide fleet, so the effect on my estimates is smaller still.

More important is how generation varies across projects. Figure 3 shows that project-level capacity factors carry information that a town or statewide average would miss. The standard deviation across projects is 0.012 around a mean of 0.143, so more than half of projects (53%) differ from the statewide mean by more than 5% and 18% differ by more than 10%. About 91% of the variance is within towns, which means that assigning each project its town average would discard most of the variation in subsidy value across roofs.

Errors in modeled generation that vary across projects pull the pass-through estimate toward one, not toward zero, because the measured subsidy enters both sides of the regression.[14] Errors in the inputs, such as a misrecorded tilt or azimuth or a geocoding error, are of this kind. Losses that PVWatts does not model and that are unrelated to its inputs, such as shading from a neighbor's tree, are not: if true generation equals modeled generation plus an independent error, the estimate is unbiased.

3.4 Subsidy Calculation

I first calculate the net present value of the subsidy for SMART projects, assuming a discount rate of 8%. Because the SMART compensation rate is locked in for ten years at the time of application, the owner's compensation in a given year depends only on that year's generation. De Groote and Verboven (2019) find that system owners heavily discount the future cash flows from such subsidy programs, so I also report results for other discount rates. I follow a standard discounted cash flow model:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{r_i \times g_i (1-a)^t}{(1+d)^t}$$

[14] Write the gross price as $P_i = c_i + (1-\rho)S_i$ and the measured subsidy as $\hat{S}_i = S_i + u_i$. The dependent variable is the net price $P_i - \hat{S}_i$, so with classical error the estimated pass-through is $\hat{\rho} = \lambda\rho + (1-\lambda)$, where $\lambda = \text{Var}(S)/[\text{Var}(S) + \text{Var}(u)]$. Equivalently, in a regression of the gross price on the measured subsidy, the coefficient $1-\rho$ is attenuated toward zero.

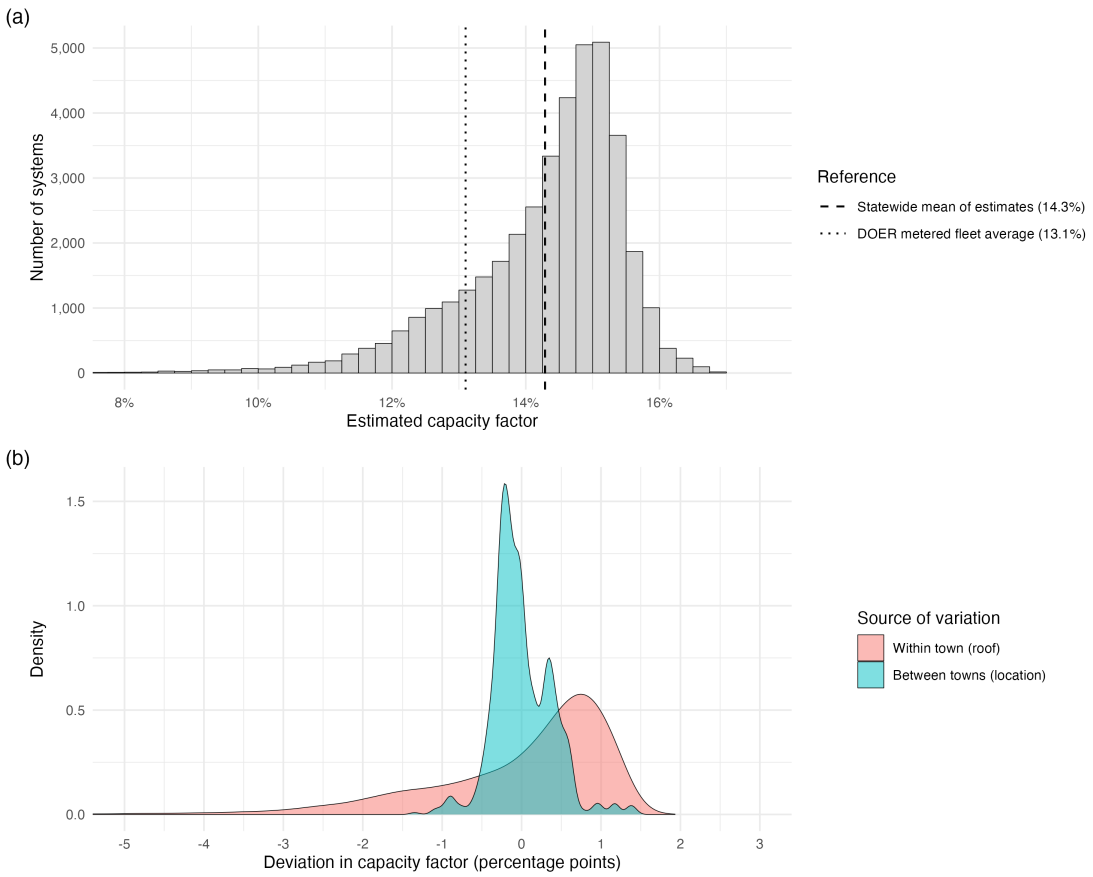


Figure 3: Variation in Modeled Capacity Factor Across Projects

where r_i is the SMART incentive rate, g_i is modeled annual generation, d is the discount rate, and a is the panel degradation rate. I assume annual degradation of 0.5% to account for the performance loss from panel aging. Because r_i and g_i do not change over the ten years, the formula reduces to $\text{NPV}_i = r_i\, g_i\, A(d)$, where $A(d) = \sum_{t=1}^{10}(1-a)^t/(1+d)^t$ is the same for every project. The discount rate therefore rescales every SMART subsidy by the same factor, which is relevant for interpreting the robustness checks in Appendix B.5.

Because SREC compensation rates were market-determined, they are not listed in the Solar Carve-Out data, and the price an owner would receive was not known at the time of purchase. In the initial years, market prices for SRECs were well below the ACP and even consistently below the auction price, so using either the ACP or the auction price alone to proxy for the compensation rate might misstate it. In the baseline, I use the midpoint of the

ACP and the auction price in each year, which is the price a buyer would expect without a view on where the market would settle between the published ACP and auction prices. As an alternative, I use the annual average market price of SRECs, which is the compensation owners actually receive and the price they would expect under rational expectations. For Solar Carve-Out II projects that were still receiving SRECs as of 2025,[15] this alternative uses the average 2024 price for the remaining years. Appendix B.5 compares these and other price assumptions.

For Solar Carve-Out projects, I apply the same approach with certificate prices:

$$\text{NPV}_i = \sum_{t=1}^{10} \frac{g_i \times f_i \times 90\% \times P_t(1-a)^t}{(1+d)^t}$$

where g_i is modeled annual generation in MWh, f_i is the project's SREC factor, d is the discount rate, a is the solar degradation rate, P_t is the SREC price in year t, and the 90% term reflects the 10% aggregator fee. I discount the ten years of payments after commercial operation for both programs, and I use the midpoint of the ACP and the auction price as the baseline price path (Appendix B.5).[16]

3.5 Summary Statistics

Table 3 presents summary statistics for installations under the Solar Carve-Out program (Panel A) and the SMART program (Panel B). The largest difference between the programs is in the subsidy per kW. Solar Carve-Out installations received an average subsidy of $1,923 per kW, about 3.4 times the mean SMART subsidy of $566 per kW. As a share of the average gross price per kW, the subsidy is about 45% under Solar Carve-Out and 15% under SMART. This gap reflects the design of the two programs: Solar Carve-Out certificates were priced by a market that tracked a high ACP, whereas SMART rates fall with the value of energy

[15] Projects with an operation date after 2016 are still receiving SRECs, but their future prices are not yet known. For example, a 2018 project will keep receiving and selling SRECs until 2028, but SREC prices for 2026 through 2028 are still unknown.

[16] The Solar Carve-Out entitlement is 40 quarters, so ten years of payments covers the whole term.

and as capacity blocks fill.

Table 3: Summary Statistics for SMART and Solar Carve-Out Samples

Variable	N	Mean	Std. Dev.	Min	Max
Panel A: Solar Carve-Out Sample					
Size (kW DC)	25,491	7.96	3.20	1.10	25.00
Total Price ($)	25,491	33,376.11	13,130.15	5,175.00	138,300.00
Price per kW ($/kW)	25,491	4,303.56	963.92	2,608.75	7,847.21
Total Subsidy ($)	25,491	15,169.04	6,435.32	1,471.30	60,419.11
Subsidy per kW ($/kW)	25,491	1,922.77	336.09	757.12	2,631.60
Net Price ($)	25,491	18,207.07	8,561.03	796.19	94,096.40
Net Price per kW ($/kW)	25,491	2,380.79	915.74	183.45	6,300.17
SREC Factor	25,491	0.92	0.10	0.80	1.00
Tilt	25,491	29.39	9.13	0.00	89.00
Azimuth	25,491	181.01	41.31	5.00	356.00
N Inverter	25,491	11.86	14.37	1	280
Annual Generation	-	-	-	-	-
Capacity Factor	25,491	0.14	0.01	0.05	0.17
Panel B: SMART Sample					
AC (kW)	14,317	7.49	2.64	1.20	23.50
Size (kW DC)	14,317	9.22	3.53	1.28	24.96
Total Price ($)	14,317	34,305.30	14,175.21	5,517.00	122,655.00
Price per kW ($/kW)	14,317	3,795.74	881.11	1,998.32	7,008.21
Total Subsidy ($)	14,317	5,182.54	3,705.16	0.00	31,917.29
Subsidy per kW ($/kW)	14,317	565.60	317.63	0.00	1,594.76
Net Price ($)	14,317	29,122.77	13,216.33	3,118.39	111,983.60
Net Price per kW ($/kW)	14,317	3,230.14	970.62	834.28	6,976.19
SMART Rate ($/kWh)	14,317	0.07	0.04	0.00	0.20
Tilt	14,317	29.40	9.48	0	84
Azimuth	14,317	179.86	56.90	0	359
N Inverter	14,317	11.57	13.52	1	78
Annual Generation	14,317	11,289.62	4,428.34	1,547.44	34,383.49
Capacity Factor	14,317	0.14	0.01	0.05	0.17

Subsidies are present values at an 8% discount rate with 0.5% annual panel degradation. Solar Carve-Out subsidies use the ACP–auction midpoint price path.

The distribution of subsidies also differs. In absolute terms the dispersion is similar (a standard deviation of $336 per kW for Solar Carve-Out and $318 per kW for SMART), but relative to the mean it is much larger under SMART (a coefficient of variation of 0.56 compared with 0.17). Every Solar Carve-Out system received at least $757 per kW, while

some SMART systems received nothing because the value of energy had caught up with the base rate. The maximum subsidy is $2,632 per kW under Solar Carve-Out and $1,595 per kW under SMART, so the two distributions barely overlap. As a result, a comparison of the two programs is a comparison of very different parts of the subsidy range.

Despite the large difference in subsidy size, both programs supported installations with similar technical characteristics. Average system sizes are comparable: 7.96 kW DC under Solar Carve-Out and 9.22 kW DC under SMART. The mean capacity factor is 0.14 in both samples. The Solar Carve-Out sample has 25,491 systems and the SMART sample 14,317.

4 Empirical Strategy

4.1 The Model

I estimate the pass-through of solar subsidies to net installation prices with the regression

$$y_{it} = \alpha + \beta \text{Subsidy}_{it} + X_{it}\gamma + \delta_{u(i)} + \delta_t + \varepsilon_{it}$$

where y_{it} is the net installation price per kW after subsidy (gross price minus subsidy, per kW of DC capacity), Subsidy_{it} is the net present value per kW of the subsidy accumulated over ten years, X_{it} is a set of controls, $\delta_{u(i)}$ is the fixed effect for the utility $u(i)$ that serves project i, and δ_t is the year fixed effect. Because y is net of the subsidy, the coefficient β equals the negative of the pass-through rate ρ: a coefficient of -1 means that a dollar of subsidy lowers the net price by a dollar (the gross price does not change), and a coefficient of -0.76 means the gross price rises by 24 cents.

The controls X_{it} include fixed effects for panel manufacturer, inverter manufacturer, installer, and city, all of which affect system prices. Panel and inverter manufacturer fixed effects control for equipment quality, installer fixed effects control for labor costs and installer quality, and city fixed effects control for demographic characteristics and local price levels.

Because SMART rates vary across utility territories, utility fixed effects allow me to identify pass-through from changes in the subsidy within a utility territory, where the compensation structure is the same. Year fixed effects capture the effect of technology and learning by doing on installation costs, which are higher on average under Solar Carve-Out than under SMART. This model therefore identifies pass-through from changes in the ACP and auction prices, which affect the market price of SRECs, from changes in the SMART compensation rate as capacity blocks fill, and from differences in generation across roofs.

I estimate the model separately for the SMART and Solar Carve-Out samples, using the net present value of the subsidy at a discount rate of 8%. The programs differ in period, in how the subsidy is measured, and in the fixed effects the policy requires, so I let every coefficient differ by program and compare the estimates directly (Appendix B.6 compares this with pooled estimation).[17] All prices and subsidies are in nominal dollars.[18]

Preferred specification. The model above uses all of the variation in the subsidy, including its movement over time, and I refer to it as the baseline. The preferred specification removes that movement. It replaces year fixed effects with year-month fixed effects (utility-by-year-month for SMART), which absorb the SREC price path and the timing of SMART block changes, adds installer-by-year and city-by-year fixed effects, and controls for the log of system size and five bins of the DC-to-AC ratio. Pass-through is then identified only from differences in the subsidy across roofs installed by the same installer in the same town and year, which come from differences in modeled generation. Section 4.2 explains why this is the cleaner source of variation.

[17] I do not include utility fixed effects for the Solar Carve-Out sample because SREC prices do not depend on the utility. Adding them leaves the Solar Carve-Out estimate essentially unchanged (Appendix Table 10).

[18] I use nominal dollars for two reasons. First, both are nominal contracts: a SMART rate is fixed in dollars per kWh at approval, SRECs are sold at dollar prices, and a buyer compares the dollar price quoted at purchase with the present value of nominal payments discounted at a nominal rate. Pass-through is a within-period object, the effect of a dollar of subsidy on the price paid on the same date, so there is no economic reason to convert to a common base year. Second, year fixed effects already absorb the economy-wide price level across years. The choice does affect the SMART estimate, because consumer prices rose by about 22% over the SMART sample while rates were stepping down. Appendix B.2 reports the estimates under alternative treatments of the price level and discusses what they imply.

4.2 Identification and Inference

The regression compares projects that face different subsidies per kW but are otherwise similar in installer, town, equipment, and year. It uses two sources of variation, which differ in how plausibly they are independent of installer pricing decisions. Each can be isolated by a different specification, and the preferred specification uses only the second.

Generation variation. Subsidy per kW depends on the electricity a system will produce, so two otherwise identical projects in different locations or with different orientations face different subsidies. This variation exists in both programs and does not depend on the policy calendar, so it cannot move together with falling installation costs over time, which makes it the cleanest source. An installer may well charge a buyer with a south-facing roof more because that buyer receives a larger subsidy, but that is the incidence the regression measures, not a threat to it. The threat is instead that roof characteristics that raise generation also change installation costs, which the controls for system size and the DC-to-AC ratio in Section 5.1 partly address. Appendix D discusses what pricing by roof implies for the interpretation of the cross-roof estimates. This source is also the one most exposed to measurement error, because PVWatts does not observe shading, and classical errors in its inputs pull the estimate toward one (Section 3.3). Section 5.1 isolates this variation by absorbing the timing of the subsidy and comparing projects of the same installer in the same town and year.

Policy variation. Under Solar Carve-Out, the subsidy per kW varies across installation years through the SREC price path and, in January 2017, through the change in the SREC factor. The year fixed effects absorb any level shift common to all projects in a year, so identification comes from how the subsidy differs across projects within a year, which reflects generation potential and the SREC factor, and from the interaction between the annual price path and the timing of payments (a project installed in 2016 receives a different sequence of vintage prices than one installed in 2018). Under SMART, the base rate falls as capacity blocks fill within a utility territory, and the value of energy differs by utility and over time.

Utility fixed effects hold the compensation schedule constant, so the variation comes from movement along the schedule. Year-month fixed effects (utility-by-year-month for SMART) absorb this variation, and the designs in Appendix B.7 isolate its discrete, rule-based part.

There are four main threats to a causal interpretation. First, installation costs fall over time, and block changes and price changes coincide with those declines. Year fixed effects absorb a common trend, but not differences in the pace of cost decline across installers or regions. Second, demand shifts can move both prices and the position on the SMART block schedule, because blocks fill faster when demand is strong. This would bias the coefficient toward zero and thus make pass-through appear lower, not higher. Third, the discount rate enters both the dependent variable and the regressor, so the coefficient mechanically depends on it. I therefore report a range of rates, although for SMART the discount rate only rescales the subsidy (Appendix B.5). Fourth, because installers know the subsidy schedule, one cannot separate a pass-through effect from installer pricing that anticipates the schedule. The preferred specification addresses the first two threats by removing the variation that comes from the timing of the subsidy, and Appendix B.7 addresses them with two designs that rely on discrete, rule-based changes in the subsidy: the cut in the SREC factor on January 8, 2017, and the step-downs in SMART rates.

Inference. I cluster standard errors at the installer level. Installers set prices, the same installer appears in many projects and years, and an installer's pricing errors are likely to be correlated across its projects. There are 385 installers in the Solar Carve-Out sample and 267 in the SMART sample, which is enough for standard asymptotic inference.[19] To compare the programs, I estimate both in one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, and I cluster by installer across programs. The point estimates equal those from separate regressions, and the standard error of the difference accounts for the 83 installers that operate in both programs, who installed

[19] Clustering at the utility level is not appropriate: with only four utility groups in the SMART sample the cluster count is too small, and in the Solar Carve-Out sample there is no utility-based policy variation. Appendix Table 11 shows how the standard errors change under alternative clustering choices. The installer-clustered errors are larger than those clustered by town in both samples, and the conclusions do not change.

56% of Solar Carve-Out projects and 69% of SMART projects.

5 Pass-Through under the Two Programs

5.1 Main Estimates

Table 4 reports pass-through estimates from specifications that progressively remove the variation in the subsidy that comes from its timing. Column (1) is the baseline. Column (2) replaces year fixed effects with year-month fixed effects, interacted with utility for SMART, which absorb the SREC price path and the timing of SMART block changes. Column (3) adds installer-by-year and city-by-year fixed effects. Column (4) adds the log of system size in kW DC and indicators for five bins of the DC-to-AC ratio. Column (5) instruments the subsidy with the project's modeled capacity factor in the specification of column (4), so that only the variation in the subsidy that comes from the roof's orientation, tilt, and location identifies pass-through.[20]

In the preferred specification, column (4), pass-through is 0.86 (0.04) under Solar Carve-Out and 0.85 (0.07) under SMART. The difference between the programs is −0.02 with a standard error of 0.085. Both estimates are high but significantly below one. Buyers capture about 85 cents of each dollar of subsidy, whether the subsidy is paid through certificates sold at an uncertain market price or through a fixed rate known at approval. Columns (3) and (5) give the same picture, with differences of 0.01 and 0.04.[21]

The cross-roof estimates rely on the variation most exposed to measurement error in generation, and classical error in the PVWatts inputs pulls the estimate toward one (Section 3.3). The pull is stronger here than in the baseline: within the same installer, town,

[20] The first-stage Wald statistics are 3,661 (Solar Carve-Out) and 1,298 (SMART). Columns (4) and (5) drop 1,988 Solar Carve-Out projects with missing AC capacity.

[21] The estimates are close to prior estimates for solar rebate programs: 78% for host-owned systems in the California Solar Initiative (CSI) (Pless and van Benthem, 2019) and 84% for Connecticut's rooftop solar rebate program (Gillingham and Tsvetanov, 2019), while Dong, Wiser and Rai (2018) document nearly complete pass-through, also under the CSI.

Table 4: Pass-Through from Cross-Roof Variation in Generation

	Dependent variable: net price per kW				
	(1)	(2)	(3)	(4)	(5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.756***	−0.799***	−0.826***	−0.863***	−0.856***
	(0.044)	(0.046)	(0.044)	(0.044)	(0.050)
Panel B: SMART					
Subsidy per kW	−1.017***	−0.887***	−0.839***	−0.847***	−0.894***
	(0.065)	(0.062)	(0.070)	(0.073)	(0.115)
Difference in pass-through (SMART − Solar Carve-Out)	0.260***	0.087	0.013	−0.016	0.038
	(0.090)	(0.080)	(0.080)	(0.085)	(0.114)
Time fixed effects	Year	Year-month	Year-month	Year-month	Year-month
Installer × year, city × year			✓	✓	✓
System size, DC/AC ratio				✓	✓
Subsidy instrumented with capacity factor					✓

Panels A and B report the coefficient on subsidy per kW. Pass-through is minus this coefficient. The row below the panels reports the difference in pass-through between the programs. All columns include installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART), or their interactions as listed. Discount rate 8%. Observations: 25,491 Solar Carve-Out (23,503 in columns 4 and 5) and 14,317 SMART. Standard errors clustered by installer in parentheses. The difference and its standard error come from one stacked regression with program-specific slopes, controls, and fixed effects, clustered by installer across programs. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

and year, roofs differ less in true generation than across the whole sample, so errors make up a larger share of the remaining variation. Instrumenting the subsidy with the modeled capacity factor (column 5) does not remove this, because the instrument comes from the same PVWatts run. If anything, then, true pass-through is below the cross-roof estimates rather than above them, which strengthens the conclusion that it is incomplete. Because both subsidies are built from the same PVWatts model, the error has the same source in both programs. Section 6.3 shows how the bound on market power changes with it.

5.2 Where the Baseline Gap Comes From

Column (1) of Table 4 is the baseline. It gives pass-through of 0.76 (0.04) under Solar Carve-Out and 1.02 (0.07) under SMART, a gap of 0.26 (0.09). Complete pass-through is rejected for Solar Carve-Out ($p < 0.001$) but not for SMART ($p = 0.80$). These estimates are robust to the choice of controls: Appendix Table 8 shows that adding equipment, installer, and city fixed effects moves them little. Taken at face value, the baseline would suggest that

a fixed-rate subsidy is passed through more fully than a certificate subsidy.

The SMART base rate is not tied to installation costs: it steps down on a fixed schedule as cumulative capacity fills the blocks, and the incentive also falls when the value of energy rises. Because blocks fill over time, however, the subsidy declines during the same years in which installation costs fall, and the SREC price path co-moves with market conditions. When subsidies and costs are correlated over time, even without a causal link, the coefficient on the subsidy reflects both how the subsidy is shared and how costs and the subsidy move together.[22] Year fixed effects remove common trends but not differences in their timing across installers. The variation in generation across roofs does not depend on the policy calendar and so avoids this problem. The year-month fixed effects alone do most of the work: the gap is already down to 0.09 (0.08) in column (2), before any of the other controls are added. As the timing of the subsidy is absorbed, the Solar Carve-Out estimate rises from 0.76 to 0.86 and the SMART estimate falls from 1.02 to 0.84–0.89. Most of the baseline gap therefore reflects how each subsidy moved over time relative to costs, not how the two designs divide the subsidy.

Three other results point the same way for SMART. First, several SMART estimates that use the timing of the subsidy are above one: 1.55 (0.23) and 1.32 (0.13) in the two lowest quartiles of the subsidy (Appendix Table 21) and 1.27 (0.15) at the step-downs in SMART rates (Appendix B.7). Over-shifting is implausible in this market (Section 6), so these estimates more likely reflect timing variation that is confounded with cost trends. Second, the baseline SMART estimate is sensitive to the treatment of the price level, because consumer prices rose by about 22% over the SMART sample while rates were stepping down. It is 1.02 in nominal dollars, 0.91 with the CPI as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars, while the Solar Carve-Out estimate stays between 0.73 and 0.80 (Appendix B.2). Every alternative moves the SMART estimate toward the

[22] Fabra and Reguant (2014) report 86% pass-through of emission costs in Spanish electricity markets and show that high pass-through can arise from cost shocks common to all firms. By the same argument, a coefficient near one here could reflect common cost movements.

cross-roof estimate. Third, two designs that use discrete, rule-based changes in the subsidy give estimates similar in magnitude to the baseline but less precise: a regression discontinuity around the January 2017 cut in the SREC factor gives Solar Carve-Out pass-through of 0.71, and a stacked event study around the step-downs in SMART rates gives 1.27 (0.15) (Appendix B.7).

Other explanations for the baseline gap. Beyond the timing of the subsidy, Appendix C considers three further explanations: SMART subsidies are much smaller, SMART rates are more predictable, and competition may have differed between the two periods. The data give little support to the third: statewide installer concentration is low throughout, most of its decline happens before SMART begins, and SMART pass-through does not vary significantly with local concentration. The first two are consistent with the data but cannot be separated from each other, because the programs differ in size, certainty, and period at once. The equal cross-roof estimates suggest that neither effect is large.

5.3 What the Comparison Rules Out

A finding of no difference is informative only to the extent that it excludes meaningful differences. In the preferred specification, the 95% confidence interval for the difference between the programs is −0.18 to 0.15. It excludes the baseline gap of 0.26, so the cross-roof estimates reject the difference that a comparison of the two programs over time would suggest, and they rule out differences in pass-through larger than about 0.18 in either direction. Column (3) gives a similar interval (−0.14 to 0.17). The instrumental variables estimates in column (5) are less precise, and their interval (−0.19 to 0.26) reaches the baseline gap.[23]

The programs differ in the form of the payment, in its predictability, and in its size (Section 2). Each difference could plausibly change pass-through, and none of them changes it by an amount the data can detect. A buyer receives about the same share of a dollar of

[23] Accounting for installers that operate in both programs barely changes the standard error of the difference in the preferred specification (0.085, compared with 0.084 if the programs were independent) but raises it in the baseline, from 0.078 to 0.090. The baseline gap remains significant ($p = 0.004$).

subsidy whether it arrives through a certificate market or through a fixed rate, and whether the subsidy covers 45% or 15% of the price. Because the two estimates are nearly identical, a single rate summarizes both programs: one slope with program-specific fixed effects gives 0.86 (0.04). I nonetheless estimate the programs separately. A pooled slope is a variance-weighted average, with about three-quarters of the weight on Solar Carve-Out, and it imposes the equality that the comparison is meant to test. In the baseline that equality is rejected, and the pooled rate of 0.84 describes neither program. Appendix B.6 reports the pooled estimates.

5.4 Robustness

Appendix B.5 reports further checks. The Solar Carve-Out estimate barely moves with the assumed SREC price path, from 0.723 with the auction price to 0.782 with the ACP. It falls from 0.805 to 0.682 as the discount rate rises from 3% to 15% and stays significantly below one at every rate, whereas the SMART estimate cannot respond to the discount rate by construction. Adding utility fixed effects to the Solar Carve-Out model leaves the estimate essentially unchanged (Appendix Table 10), and clustering by town or utility instead of by installer does not change the conclusions (Appendix Table 11).

6 What Pass-Through Reveals about Market Power

Pless and van Benthem (2019) formalize pass-through over-shifting as a test for market power: under perfect competition with non-decreasing marginal cost, pass-through cannot exceed one, so more-than-complete pass-through, together with sufficiently convex demand, reveals that firms have market power. This section uses the same framework in the opposite direction. When demand is log-concave, market power pushes pass-through below one, so a pass-through rate near one, as found in this market, limits how much market power there can be. The cross-roof variation of Section 5.1 is well suited to this question, because it

removes the supply side from the pass-through formula. I first set out the framework and estimate demand, then derive the bound for SMART and the division of surplus it implies, and finally compare them with reduced-form evidence.

6.1 Framework

Weyl and Fabinger (2013) provide a framework for interpreting the size of pass-through. Consider a symmetric market in which installers' conduct is summarized by a parameter θ, which equals 0 under perfect competition and 1 under monopoly and does not change with the subsidy. The pass-through rate of a per-unit subsidy is

$$\rho = \frac{1}{1 + \frac{\varepsilon_D - \theta}{\varepsilon_S} + \frac{\theta}{\varepsilon_{ms}}}, \tag{1}$$

where ε_D is the absolute value of the demand elasticity, ε_S is the elasticity of supply, and ε_{ms} is the elasticity of the marginal consumer surplus, which measures the curvature of demand. The term $1/\varepsilon_{ms}$ is zero when demand is log-linear, positive when demand is less convex than log-linear (for example, linear), and negative when it is more convex (for example, constant elasticity).[24]

Pass-through equals one only when the last two terms in the denominator sum to zero, which happens in four situations: (i) perfect competition with constant marginal cost, (ii) perfect competition with perfectly inelastic demand, (iii) market power with constant marginal cost and log-linear demand, and (iv) market power with demand more convex than log-linear together with rising marginal cost, so that the two forces offset. In situation (iii), the profit-maximizing markup is a fixed dollar amount that does not respond to the subsidy. Situation (ii) is implausible for residential solar, whose demand responds to price, and (iv) requires two forces to cancel by coincidence. Situations (i) and (iii) are the leading candi-

[24] Pless and van Benthem (2019) use the same expression, with an additional term for conduct that changes with the subsidy.

dates, and they have different implications for who gains from the subsidy. The ratio of the gain in consumer surplus to the gain in producer surplus is $I = \rho/[1-(1-\theta)\rho]$, which equals $1/\theta$ when $\rho = 1$. Under (i) installers gain nothing. Under (iii) they gain because they sell more systems at a positive markup. A pass-through rate near one therefore does not by itself show that installers capture none of the subsidy.

Market-wide and roof-specific pass-through. Equation (1) describes a subsidy that changes for all buyers at once. Such a change shifts market demand and moves installers along their marginal cost curves, so it can be written as $1/\rho_M - 1 = r + \theta/\varepsilon_{ms}$, where $r \geq 0$ is the slope of marginal cost relative to the slope of inverse demand. The cross-roof estimates use a different kind of variation: differences in the subsidy across buyers of the same installer in the same town and year. A higher subsidy for one buyer does not change the installer's volume appreciably, so the supply term drops out, and the pass-through of a roof-specific subsidy is

$$\frac{1}{\rho_B} - 1 = \frac{\theta_B}{\varepsilon^B_{ms}},$$

where θ_B measures the installer's pricing power over individual buyers and ε^B_{ms} is the curvature of demand among buyers with similar roofs. Two implications follow. First, if installers charged one price per kW regardless of the roof, or if the market were perfectly competitive, then $\rho_B = 1$. A cross-roof estimate below one therefore means that the gross price rises with the roof's subsidy, either because installers price according to the value of the roof or because roofs that generate more cost more to install in ways that system size and the DC-to-AC ratio do not capture. Second, without the supply term the whole shortfall below one is attributed to conduct and curvature, so the cross-roof estimate bounds market power directly. With the same conduct and curvature, $\rho_M \leq \rho_B$, so the cross-roof estimate is also an upper bound on market-wide pass-through, which is consistent with the reading in Section 5.2 that the baseline SMART estimate of 1.02 is pushed up by the timing of the subsidy.

6.2 Demand

Distinguishing (i) from (iii) requires the curvature of demand. I estimate demand following Pless and van Benthem (2019). The unit is a ZIP code and year, and the outcome is installations per 1,000 residents. Demand is quadratic in the mean net price per watt,

$$q_{zt} = \beta_1 p_{zt} + \beta_2 p_{zt}^2 + X_z\gamma + \delta_{c(z)} + \tau_t + e_{zt}, \tag{2}$$

so that β_2 measures convexity. The vector X_z contains ZIP characteristics from the 2017–2021 American Community Survey (log median household income, the share of adults with a bachelor's degree, median age, log median home value, and the share of owner-occupied homes), $\delta_{c(z)}$ are county fixed effects, and τ_t are year fixed effects. The net price responds to demand shocks, so I instrument p and p^2 with the mean subsidy per watt and its square and with the average weekly wage in the county's building-equipment contracting sector (NAICS 2382, from the Quarterly Census of Employment and Wages) and its square. The subsidy shifts the price buyers pay, and the wage shifts installation costs. Neither should affect adoption except through price. This is the functional form that Pless and van Benthem (2019) use, with the same types of instruments, so a difference in curvature between California and Massachusetts is not an artifact of the specification. Standard functional forms impose the curvature of demand: linear demand is log-concave, exponential demand is log-linear, and constant-elasticity demand is log-convex (Fabinger and Weyl, 2014). I use a quadratic because it is the lowest-order form that estimates curvature rather than imposing it, and I evaluate curvature at the mean price because demand curves need not have constant curvature. Log-concavity is also what one would expect for a discrete adoption decision when willingness to pay has a log-concave density (Bagnoli and Bergstrom, 2005). Appendix B.10 replaces the quadratic with a natural spline. From the estimates, I compute the price elasticity and the curvature $1/\varepsilon_{ms} = 1 - qq''/q'^2$ at the mean price, with standard errors by the delta method. Because the quadratic in levels imposes a particular shape, I

also estimate the same equation with $\log q_{zt}$ as the outcome. In that form the coefficient on p^2 is zero exactly when demand is log-linear, so it is a direct test of situation (iii), and $1/\varepsilon_{ms} = -2\beta_2/(\beta_1 + 2\beta_2 p)^2$. The sample includes ZIP-years with at least one installation and at least 500 residents.

Table 5: Demand Estimates

	Panel A: Solar Carve-Out		Panel B: SMART	
Dependent variable	Quadratic q (1)	Log-quadratic $\log q$ (2)	Quadratic q (3)	Log-quadratic $\log q$ (4)
Net price	0.777	2.110	2.516***	2.343***
	(4.257)	(2.690)	(0.605)	(0.563)
Net price squared	−0.124	−0.362	−0.439***	−0.446***
	(0.716)	(0.459)	(0.082)	(0.075)
Price elasticity at the mean	0.27	0.79	−1.57***	−2.16***
	(1.21)	(1.09)	(0.42)	(0.42)
Curvature $1/\varepsilon_{ms}$ at the mean	14.86	7.05	5.38*	2.14**
	(56.01)	(11.35)	(2.77)	(1.02)
First-stage Wald statistic, p / p^2	20.3 / 16.3	20.3 / 16.3	61.2 / 56.3	61.2 / 56.3
Mean net price ($/W)	2.47	2.47	3.35	3.35
Mean installations per 1,000 residents	1.54	1.54	0.91	0.91
ZIP-years	2,171	2,171	1,715	1,715

Two-stage least squares estimates. Columns (1) and (3): equation (2), with q the installations per 1,000 residents. Columns (2) and (4): the same equation with $\log q$ as the outcome, so that the coefficient on the squared price is zero under log-linear demand. Instruments: mean subsidy per watt and its square, and the county building-equipment contractor wage and its square. Controls: ACS 2017–2021 ZIP characteristics, county and year fixed effects. $1/\varepsilon_{ms} = 0$ for log-linear demand, > 0 for log-concave demand, and < 0 for log-convex demand. Standard errors clustered by ZIP code in parentheses. Elasticity and curvature standard errors by the delta method. Appendix B.9 reports alternative specifications. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 5 reports the estimates. Under SMART, demand slopes down with an elasticity of −1.57 (standard error 0.42) at the mean price of $3.35 per watt. This is nearly twice the −0.85 that Pless and van Benthem (2019) report for host-owned systems in California at a mean price of $3.70 per watt.[25] The squared term is negative and precisely estimated, so demand is concave, and the curvature measure is 5.38 (2.77). The log-quadratic form

[25]The difference may reflect a later period in which buyers had more alternatives to buying a system outright, such as leases and power purchase agreements, which are outside my sample. It may also reflect who is buying: early adopters are more likely to install at almost any price, so later buyers may simply be more price-responsive. Differences in samples and instruments may also contribute.

gives the same answer more sharply: the coefficient on the squared price is −0.446 (0.075), which rejects log-linear demand, and the curvature is 2.14 (1.02). Demand under SMART is therefore log-concave, not log-linear and not log-convex, whereas Pless and van Benthem (2019) find convex demand in California with the same functional form.[26] The result holds across specifications. The elasticity ranges from −1.14 to −1.92 and the curvature from 3.4 to 9.8 when I add utility fixed effects, replace county and year fixed effects with county-by-year fixed effects, weight by population, drop ZIP-years with fewer than three installations, or cluster by county (Appendix B.9). Under Solar Carve-Out, demand is not identified. The subsidy is driven by the statewide certificate price and varies little across ZIP codes within a county and year, and the estimates change sign across specifications. I therefore draw structural conclusions only for SMART.

Two further limitations qualify the SMART result. First, the curvature is measured with limited precision: the quadratic curvature of 5.38 has a t-statistic of about 1.9 and the log-quadratic curvature of 2.14 about 2.1, and the bound on market power below depends on its sign. Second, the sample excludes ZIP-years with no installations, which truncates demand at low quantities, and truncation affects the estimated curvature directly.

6.3 Market Power under SMART

With roof-specific variation, the bound follows directly from the cross-roof estimate. Because r is absent, $\theta_B = (1/\rho_B - 1)\,\varepsilon_{ms}$ holds with equality if all of the shortfall below one reflects pricing, and it is an upper bound if part of the shortfall reflects cost differences across roofs. At $\rho_B = 0.847$ (Table 4, column 4) and the curvature estimates in Table 5, θ is at most 0.034 with the quadratic demand and 0.085 with the log-quadratic demand. Since the Lerner index equals θ/ε_D, installer markups are at most 2–4% of price. This range, from the two quadratic forms, is the one I report throughout. The calculation uses the curvature

[26] Both forms bend back at low prices: the level quadratic slopes upward below \$2.87 per watt (a quarter of ZIP-years) and the log-quadratic below \$2.62 (13%), so the estimates describe demand around the mean rather than over the whole range.

of aggregate ZIP-level demand in place of the curvature among buyers with similar roofs. The upper percentiles are considerably wider: drawing ρ and $1/\varepsilon_{ms}$ from their estimated distributions, the 95th percentile of the bound is 0.15 with the quadratic demand and 0.35 with the log-quadratic demand, which correspond to markups of about 10 and 16 percent of price at the estimated elasticities. With a natural spline in place of the quadratic, demand remains log-concave, but the curvature at the mean is smaller, and the bound on markups rises to 4–7% of price (Appendix B.10).

Measurement error in generation pulls the cross-roof estimate toward one (Section 3.3), so true pass-through is lower and the bound wider. If a share λ of the remaining variation in the measured subsidy is true variation, then $\rho_B = 1 - (1 - \hat{\rho}_B)/\lambda$. Table 6 reports the bound for values of λ from 1 to 0.5. Markups exceed 5% of price only if λ is below 0.82 with the log-quadratic demand or 0.52 with the quadratic demand, and they exceed 10% only if about half or more of the remaining variation is error (λ below 0.48 and 0.33). The conclusion that markups are small therefore holds unless modeled generation is a poor guide to differences in true generation among roofs of the same installer, town, and year.

Table 6: Bound on Market Power under Measurement Error in Generation

		Quadratic demand		Log-quadratic demand	
λ	ρ_B	θ bound	Markup (% of price)	θ bound	Markup (% of price)
1.0	0.847	0.034	2.1	0.085	3.9
0.9	0.830	0.038	2.4	0.096	4.4
0.8	0.808	0.044	2.8	0.111	5.1
0.7	0.781	0.052	3.3	0.131	6.1
0.6	0.745	0.064	4.1	0.161	7.4
0.5	0.693	0.082	5.2	0.207	9.6
λ at which the markup reaches 5%: 0.52 (quadratic), 0.82 (log-quadratic)					
λ at which the markup reaches 10%: 0.33 (quadratic), 0.48 (log-quadratic)					

λ is the share of the variation in the measured subsidy, within the cells of the preferred specification, that reflects true differences in generation. $\rho_B = 1 - (1 - \hat{\rho}_B)/\lambda$ with $\hat{\rho}_B = 0.847$ (Table 4, column 4). The bound is $\theta \leq (1/\rho_B - 1)\,\varepsilon_{ms}$ and the markup is θ/ε_D, at the point estimates of the SMART demand curvature and elasticity in Table 5.

The market-wide bound from the baseline estimate is tighter but more exposed to the timing of the subsidy. At $\rho = 1.017$ and a positive $1/\varepsilon_{ms}$, the bound is zero. Drawing ρ and $1/\varepsilon_{ms}$ from their estimated distributions, the 95th percentile of the bound is 0.025 with the quadratic demand and 0.060 with the log-quadratic demand, and fewer than 3% of draws imply log-convex demand. These bounds imply a markup of at most about 2–3% of price.[27]

The same framework gives the division of surplus. With the incidence ratio $I = \rho/[1 - (1 - \theta)\rho]$ from Section 6.1, the buyers' share of the surplus that the subsidy creates is $I/(1 + I) = \rho/(1 + \theta\rho)$, which equals ρ under perfect competition. At the cross-roof estimate ($\rho = 0.847$), buyers receive 85% of the surplus if the market is perfectly competitive and 82% and 79% at the bounds on θ of 0.034 and 0.085. Market power at the bound therefore shifts at most a few percentage points of the surplus from buyers to installers.

The SMART evidence therefore points to situation (i) or close to it: a nearly competitive market in which marginal cost is close to flat, so that buyers receive most of the subsidy and installers gain little from it.[28] The bound assumes that conduct does not change with the subsidy and that marginal cost does not fall with volume. Economies of scale would loosen it. Appendix D discusses pricing by roof further, including its welfare effects and two tests.

The comparison with Pless and van Benthem (2019) is direct, because the demand specification is the same. In California, pass-through of 153% to lessees together with convex demand revealed market power in the third-party-owned segment, while host-owned buyers received 78% of the subsidy. In Massachusetts, host-owned buyers receive about 85% of the subsidy under both programs, and log-concave demand turns high pass-through into an upper bound on market power rather than evidence of it. Over-shifting can reveal market power. High pass-through with log-concave demand means the opposite. The two results are not in conflict, since they concern different segments, periods, and states, and the bound

[27] The same equation also bounds the supply slope: $r \leq 1/\rho - 1$, which is at most about 0.12 at the lower end of the confidence interval for the baseline ρ.

[28] The SMART estimate of 1.02, with a 95% confidence interval of about 0.89 to 1.14, is not evidence of over-shifting. The interval includes values below one, and over-shifting requires market power together with log-convex demand (Pless and van Benthem, 2019), whereas the estimated demand is log-concave.

here applies to host-owned systems under SMART.

6.4 Supporting Evidence

The bound rests on two assumptions that the data can partly check: that marginal cost does not rise steeply with volume, and that conduct does not differ with local competition. If either failed, the reduced-form patterns would show it, so Table 7 tests the predictions of their failure, with the aim of rejecting them. First, a rising marginal cost implies lower pass-through when installers are busy. Second, market power implies higher gross prices where installers face fewer competitors. Because city fixed effects are included, this comparison is within a town over time. Third, if conduct differs across towns, pass-through varies with concentration.[29]

These checks agree with the structural bound for SMART. Pass-through does not fall when an installer is busy (0.044 per standard deviation of workload, standard error 0.023). Statewide workload lowers it by 0.052 (0.022) per standard deviation, from 1.03 at mean volume to about 0.98 one standard deviation above it, which is the small, market-wide supply response that the bound on r allows. Gross prices are lower where more installers are active, by \$66 per kW per log point ($p = 0.05$), or 1.7% of the mean gross price of \$3,796 per kW. This is small, and it may also reflect demand, since installers enter towns where demand is growing. Pass-through rises with concentration by 0.072 (0.043) per standard deviation, which is not significant at the 5% level. These are correlations rather than causal estimates, because workload and the number of competitors respond to demand. The same pattern appears when SMART pass-through is estimated by tercile of ZIP-level installer concentration: 0.96, 1.04, and 1.06 from the least to the most concentrated tercile, and equality is not rejected ($p = 0.108$, Appendix B.8). If conduct differed across towns, pass-through would be lower

[29] I measure an installer's workload as the log of its installations in the month of operation and the month before, relative to the installer's own mean, and statewide workload as the log of monthly installations relative to the mean of that year, and I interact each, standardized, with the subsidy. For town competition, I regress the gross price per kW on the log number of installers active in the town and year and, separately, on the town-year HHI, for towns with at least ten installations.

Table 7: Workload, Competition, and Pass-Through

	Net price per kW			Gross price per kW	
	Installer workload (1)	Statewide workload (2)	Town HHI (3)	Town competitors (4)	Town HHI (5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.757***	−0.758***	−0.779***		
	(0.044)	(0.044)	(0.039)		
Subsidy per kW × installer workload	−0.010				
	(0.020)				
Subsidy per kW × statewide workload		0.003			
		(0.029)			
Subsidy per kW × town HHI			−0.052*		
			(0.030)		
Log number of installers in town-year				42.5	
				(31.3)	
Town HHI					−95.7***
					(14.4)
Observations	25,491	25,491	21,783	21,783	21,783
Panel B: SMART					
Subsidy per kW	−1.019***	−1.027***	−1.035***		
	(0.065)	(0.064)	(0.070)		
Subsidy per kW × installer workload	−0.044*				
	(0.023)				
Subsidy per kW × statewide workload		0.052**			
		(0.022)			
Subsidy per kW × town HHI			−0.072*		
			(0.043)		
Log number of installers in town-year				−66.1*	
				(34.0)	
Town HHI					−0.8
					(16.3)
Observations	14,317	14,317	11,346	11,346	11,346

The dependent variable is the net price per kW in columns (1)–(3) and the gross price per kW in columns (4) and (5). Installer workload is the log of the installer's installations in the month of operation and the month before, minus the installer's mean. Statewide workload is the log of all installations in the month, minus the mean for that year. Town competition is measured in town-years with at least ten installations, and the HHI is on a 0–10,000 scale. Workload and HHI are standardized within each program. The coefficient on subsidy per kW is minus pass-through at the mean, and the coefficient on an interaction is minus the change in pass-through per standard deviation. Gross-price regressions also control for subsidy per kW. All regressions include the full set of fixed effects (including city and year). Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

where competition is weaker.

6.5 Solar Carve-Out

The two programs pass through the same share of the subsidy across roofs (Section 5.3), and 83 installers, accounting for 56% of Solar Carve-Out projects, operate in both. If conduct and the curvature of demand were similar in the two periods, the SMART bound would apply to Solar Carve-Out as well. The data cannot confirm this, because demand under Solar Carve-Out is not identified. Without a demand estimate, the Solar Carve-Out results rest on the reduced-form checks, and they do not fit a rising supply curve. Under perfect competition, a pass-through estimate of 0.76 would imply a supply elasticity about three times the demand elasticity (about six times at the cross-roof estimate of 0.86), but pass-through does not change with installer workload (0.010, standard error 0.020) or statewide workload (−0.003, 0.029). The number of competitors in a town has no relation to gross prices, and prices are lower, not higher, in concentrated town-years (−$96 per kW per standard deviation). A likely reason is the Solarize Mass campaigns, in which a town selected a single installer and offered tiered group prices that fell as more households signed up, which made a town concentrated and cheap at the same time (Clean Energy States Alliance, 2015).[30] The Solar Carve-Out estimate is close to the 0.78 that Pless and van Benthem (2019) find for host-owned systems in California. None of the forces in equation (1) that I can measure accounts for the shortfall below one. The estimate measures pass-through relative to the present value of the subsidy at an 8% discount rate. If households value uncertain certificate revenue at less than that, the measured rate falls below one even when installers pass on the whole of what buyers value. This points to the size and predictability of the subsidy, discussed in Appendix C.1.

[30]For example, Lexington and Bedford selected Astrum Solar in a campaign that ran until June 2014 (Town of Lexington, 2014), and Astrum, as Direct Energy Solar, is the second-largest installer in the 2015 data.

7 Conclusion

I estimate the pass-through of two production-based solar subsidies in Massachusetts, using subsidy measures built from each roof's modeled generation. The paper has two findings.

First, the two designs pass through the subsidy at the same rate. When only differences in generation across roofs identify pass-through, it is about 85% under both the certificate-based Solar Carve-Out and the fixed-rate SMART program, and the data rule out differences larger than about 0.18. The gap between the baseline estimates of 76% and 102% comes from the timing of each subsidy relative to falling installation costs. Pass-through also barely depends on the assumed SREC prices, does not fall when installers are busy or where they are more concentrated, and shows no robust income gradient within SMART.

Second, high pass-through bounds market power. Because demand under SMART is log-concave and variation across roofs removes the supply side, the shortfall below one limits installer markups to a few percent of price at the point estimates, and to about 5% even with substantial measurement error in generation. Unlike California, where convex demand and over-shifting to lessees indicated market power, the market for host-owned systems under SMART appears close to competitive.

Three lessons follow for policy and for research on subsidy design. Comparisons of designs across programs or periods confound the design with the cost trends that accompany it, so they should rely on variation that does not move with the policy calendar. In a competitive installer market, a production-based subsidy reaches buyers largely, though not fully, whether it is paid through a certificate market or a fixed rate, so the choice between the two can rest on other considerations. Finally, most of the variation in subsidy value is across roofs within the same town, so measures that assign town or statewide averages discard it, and errors in measured generation pull pass-through estimates toward one.

References

Aguirre, Iñaki, Simon Cowan, and John Vickers. 2010. "Monopoly price discrimination and demand curvature." *American Economic Review*, 100(4): 1601–1615.

Bagnoli, Mark, and Ted Bergstrom. 2005. "Log-Concave Probability and Its Applications." *Economic Theory*, 26(2): 445–469.

Barwick, Panle Jia, Hyuk-soo Kwon, Binglin Wang, and Nahim Bin Zahur. 2023. "Pass-through of electric vehicle subsidies: A global analysis." Vol. 113, 323–328.

Bielen, David A., Eric J. O'Shaughnessy, Benjamin O. Sigrin, and Robert M. Margolis. 2019. "Value Accrual to Customers, Installers, and Financiers in Third-Party Owned Solar PV Markets." National Renewable Energy Laboratory NREL/TP-6A20-70552, Golden, CO.

Borenstein, Severin. 2017. "Private net benefits of residential solar PV: The role of electricity tariffs, tax incentives, and rebates." *Journal of the Association of Environmental and Resource Economists*, 4(S1): S85–S122.

Burr, Chrystie. 2016. "Subsidies and investments in the solar power market." *University of Colorado at Boulder Working Paper*.

Busse, Meghan, Jorge Silva-Risso, and Florian Zettelmeyer. 2006. "$1,000 cash back: The pass-through of auto manufacturer promotions." *American Economic Review*, 96(4): 1253–1270.

Butters, R Andrew, Daniel W Sacks, and Boyoung Seo. 2022. "How do national firms respond to local cost shocks?" *American Economic Review*, 112(5): 1737–1772.

Cabral, Marika, Michael Geruso, and Neale Mahoney. 2018. "Do larger health insurance subsidies benefit patients or producers? Evidence from Medicare Advantage." *American Economic Review*, 108(8): 2048–2087.

Clean Energy States Alliance. 2015. "States Spur Solarize Successes." *https://www.cesa.org/states-spur-solarize-successes/*.

Colas, Mark, and Emmett Reynier. 2026. "Optimal Subsidies for Residential Solar." *Journal of Political Economy Microeconomics*, 4(1): 213–264.

Crago, Christine Lasco, and Ilya Chernyakhovskiy. 2017. "Are policy incentives for solar power effective? Evidence from residential installations in the Northeast." *Journal of Environmental Economics and Management*, 81: 132–151.

De Groote, Olivier, and Frank Verboven. 2019. "Subsidies and time discounting in new technology adoption: Evidence from solar photovoltaic systems." *American Economic Review*, 109(6): 2137–2172.

Deline, Chris, Dirk Jordan, Bill Sekulic, Josh Parker, Byron McDanold, and Allan Anderberg. 2024. "PV Lifetime Project-2024 NREL Annual Report." National Renewable Energy Laboratory (NREL), Golden, CO (United States).

Dong, Changgui, Ryan Wiser, and Varun Rai. 2018. "Incentive pass-through for residential solar systems in California." *Energy Economics*, 72: 154–165.

Fabinger, Michal, and E. Glen Weyl. 2014. "A Tractable Approach to Pass-Through Patterns." Working paper.

Fabra, Natalia, and Mar Reguant. 2014. "Pass-through of emissions costs in electricity markets." *American Economic Review*, 104(9): 2872–2899.

Fan, Ying, and Ge Zhang. 2022. "The welfare effect of a consumer subsidy with price ceilings: the case of Chinese cell phones." *The RAND Journal of Economics*, 53(2): 429–449.

Ganapati, Sharat, Joseph S Shapiro, and Reed Walker. 2020. "Energy cost pass-through in US manufacturing: Estimates and implications for carbon taxes." *American Economic Journal: Applied Economics*, 12(2): 303–342.

Genakos, Christos, and Mario Pagliero. 2022. "Competition and pass-through: evidence from isolated markets." *American Economic Journal: Applied Economics*, 14(4): 35–57.

Gerarden, Todd D. 2023. "Demanding innovation: The impact of consumer subsidies on solar panel production costs." *Management Science*, 69(12): 7799–7820.

Gillingham, Kenneth, and Tsvetan Tsvetanov. 2019. "Hurdles and steps: Estimating demand for solar photovoltaics." *Quantitative Economics*, 10(1): 275–310.

Gulati, Sumeet, Carol McAusland, and James M Sallee. 2017. "Tax incidence with endogenous quality and costly bargaining: Theory and evidence from hybrid vehicle subsidies." *Journal of Public Economics*, 155: 93–107.

Hughes, Jonathan E, and Molly Podolefsky. 2015. "Getting green with solar subsidies: evidence from the California solar initiative." *Journal of the Association of Environmental and Resource Economists*, 2(2): 235–275.

Hu, Yunyi, Haitao Yin, and Li Zhao. 2023. "Subsidy phase-out and consumer demand dynamics: Evidence from the battery electric vehicle market in China." *Review of Economics and Statistics*, 1–50.

Kenkel, Donald S. 2005. "Are alcohol tax hikes fully passed through to prices? Evidence from Alaska." *American Economic Review*, 95(2): 273–277.

Kirwan, Barrett E. 2009. "The incidence of US agricultural subsidies on farmland rental rates." *Journal of political economy*, 117(1): 138–164.

Kopczuk, Wojciech, Justin Marion, Erich Muehlegger, and Joel Slemrod. 2016. "Does Tax-Collection Invariance Hold? Evasion and the Pass-Through of State Diesel Taxes." *American Economic Journal: Economic Policy*, 8(2): 251–286.

Lade, Gabriel E, and James Bushnell. 2019. "Fuel subsidy pass-through and market structure: Evidence from the renewable fuel standard." *Journal of the Association of Environmental and Resource Economists*, 6(3): 563–592.

Matisoff, Daniel C, and Erik P Johnson. 2017. "The comparative effectiveness of residential solar incentives." *Energy Policy*, 108: 44–54.

Miklós-Thal, Jeanine, and Greg Shaffer. 2021. "Pass-through as an economic tool: On exogenous competition, social incidence, and price discrimination." *Journal of Political Economy*, 129(1): 323–335.

Muehlegger, Erich, David S Rapson, et al. 2018. "Subsidizing mass adoption of electric vehicles: Quasi-experimental evidence from California." National Bureau of Economic Research.

Nakamura, Emi, and Dawit Zerom. 2010. "Accounting for incomplete pass-through." *The review of economic studies*, 77(3): 1192–1230.

Newey, Whitney K., James L. Powell, and Francis Vella. 1999. "Nonparametric Estimation of Triangular Simultaneous Equations Models." *Econometrica*, 67(3): 565–603.

Pless, Jacquelyn, and Arthur A. van Benthem. 2019. "Pass-through as a test for market power: An application to solar subsidies." *American Economic Journal: Applied Economics*, 11(4): 367–401.

Podolefsky, Molly. 2013. "Tax evasion and subsidy pass-through under the solar investment tax credit." *Working Paper*.

Ros, Agustin J, and Sai Shetty Sai. 2023. "Residential rooftop solar demand in the US and the impact of net energy metering and electricity prices." *Energy Economics*, 118: 106491.

Sallee, James M. 2011. "The surprising incidence of tax credits for the Toyota Prius." *American Economic Journal: Economic Policy*, 3(2): 189–219.

Town of Lexington. 2014. "What is Solarize Lexington-Bedford?" *https://lexingtonma.gov/DocumentCenter/View/1411/Solarize-Mass-PDF*.

U.S. Energy Information Administration. 2023. "Federal Financial Interventions and Subsidies in Energy in Fiscal Years 2016–2022." *U.S. Department of Energy, Washington, DC. https://www.eia.gov/analysis/requests/subsidy/*.

Weyl, E Glen, and Michal Fabinger. 2013. "Pass-through as an economic tool: Principles of incidence under imperfect competition." *Journal of political economy*, 121(3): 528–583.

A Data Construction

Merging DOER and Tracking the Sun. Projects are identified differently in the two sources. I match them on operation date, size, and total cost simultaneously. This gives an almost perfect one-to-one match, with only a few projects matched to the same Tracking the Sun record, and I verified manually that these projects have the same technical details.

Dates. For the SMART sample, the DOER data list two dates for each project: the Commercial Operation Date and the Incentive Payment Effective Date. I use the Commercial Operation Date, as this is when compensation rates were determined.

Geocoding. The PVWatts location input accepts street addresses, ZIP codes, or coordinates. The public DOER data contain only ZIP codes, which can cover large areas relative to the resolution of the solar resource data, so I obtained detailed addresses for each project through a separate data request to DOER. The calculator returned unreliable results for street addresses entered directly, so I converted the addresses to coordinates with the U.S. Census geocoder and used the OpenStreetMap geocoding service for addresses that the Census geocoder could not match. All addresses were geocoded by one of the two services, and I verified that all coordinates lie in Massachusetts.

B Additional Results

B.1 Baseline Specifications

Table 8 reports the baseline model as controls are added. Columns (1) and (5) control only for year fixed effects in Panel A (Solar Carve-Out) and for utility and year fixed effects in Panel B (SMART). Columns (2) and (6) add inverter and panel manufacturer fixed effects. Columns (3) and (7) add installer and city fixed effects. Columns (4) and (8) include all controls and are the baseline in Table 4, column (1).

Table 8: Baseline Pass-Through Estimates: Solar Carve-Out and SMART

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Subsidy per kW	−0.709***	−0.729***	−0.751***	−0.756***	−0.908***	−0.888***	−1.005***	−1.017***
	(0.109)	(0.073)	(0.042)	(0.044)	(0.138)	(0.089)	(0.061)	(0.065)
Controls								
Inverter		✓		✓		✓		✓
Manufacturer		✓		✓		✓		✓
Installer			✓	✓			✓	✓
City			✓	✓			✓	✓
Utility FE					✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491	14,317	14,317	14,317	14,317

Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation. SREC subsidies use the midpoint of the ACP and auction prices.

Across specifications in Panel A, the pass-through estimates without controls for factors that affect system prices, and with controls for only some cost factors, are all close to the full model, which suggests that the estimated pass-through rate is not driven by variation in cost factors that the controls capture. Specifications in Panel B are less similar to one another. Adding manufacturer and inverter controls alone lowers the estimate slightly (0.888), while adding installer and city fixed effects raises it to 1.005, very close to the full model (1.017). The standard errors fall as controls are added, from 0.138 to 0.065 in Panel B, which is

consistent with the controls removing cost variation that is unrelated to the subsidy. I use columns (4) and (8) as the baseline.

B.2 Price Level

The baseline is in nominal dollars, for the reasons given in Section 4. The SMART estimate is, however, sensitive to how price-level changes are treated: it is 1.02 in nominal dollars, 0.91 when the CPI is added as a control, 0.93 with year-by-month fixed effects, and 0.85 in CPI-deflated dollars (Table 9), while the Solar Carve-Out estimate stays between 0.73 and 0.80. Consumer prices rose by about 22% over the SMART sample while rates were stepping down, so each of these alternatives also removes part of the co-movement between the subsidy and costs discussed in Section 5.2. I therefore read the baseline SMART result as pass-through that is high, between roughly 85 and 100 percent, and near but not exactly complete. The cross-roof estimate of 0.85 lies at the lower end of this range. The four SMART estimates come from the same sample, so their standard errors alone do not show whether they differ. Their 95% confidence intervals overlap (0.89–1.14 in nominal dollars and 0.72–0.97 in CPI-deflated dollars), but that does not establish that they are equal, and a formal test requires the covariance between them.

B.3 Solar Carve-Out with Utility Fixed Effects

Table 10 adds utility fixed effects to the full Solar Carve-Out model. SREC prices do not depend on the utility, so the estimate is essentially unchanged.

Table 9: Pass-Through under Alternative Treatments of the Price Level

	Dependent variable: net price per kW				
	Nominal dollars			CPI-deflated dollars	
	Year FE (baseline) (1)	Year FE, CPI control (2)	Year-by-month FE (3)	Year FE (4)	Year-by-month FE (5)
Panel A: Solar Carve-Out					
Subsidy per kW	−0.756***	−0.760***	−0.799***	−0.729***	−0.797***
	(0.044)	(0.046)	(0.046)	(0.042)	(0.045)
Panel B: SMART					
Subsidy per kW	−1.017***	−0.906***	−0.929***	−0.846***	−0.883***
	(0.065)	(0.063)	(0.062)	(0.062)	(0.062)

Full specifications with installer, inverter, manufacturer, and city fixed effects (and utility fixed effects for SMART). Pass-through is minus the coefficient on subsidy per kW. Discount rate 8%. Deflated values use the monthly CPI-U and are expressed in dollars of the first half of 2019 (index 254.412). The CPI ranges from 234 to 253 over the Solar Carve-Out sample and from 248 to 302 over the SMART sample. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 10: Solar Carve-Out Pass-Through with and without Utility Fixed Effects

	Baseline	With Utility FE
Subsidy per kW	−0.756***	−0.756***
	(0.044)	(0.044)
Installer, inverter, manufacturer, city, year FE	✓	✓
Utility FE		✓
Observations	25,491	25,328

Standard errors clustered by installer in parentheses. Discount rate 8%. The number of observations falls because singleton groups are dropped. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.4 Alternative Clustering

Table 11: Standard Errors under Alternative Clustering

Clustered by	Installer	City	Utility
Panel A: Solar Carve-Out			
Subsidy per kW	−0.756***	−0.756***	−0.756***
	(0.044)	(0.029)	(0.060)
Panel B: SMART			
Subsidy per kW	−1.017***	−1.017***	−1.017***
	(0.065)	(0.046)	(0.035)

Coefficient on subsidy per kW from the baseline specifications. The dependent variable is the net price per kW, and pass-through is minus the coefficient. Discount rate 8%. Standard errors in parentheses under each clustering. Utility clustering has only four clusters in the SMART sample and is shown for comparison only. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.5 SREC Prices, Discount Rates, and Tests of Complete Pass-Through

SREC prices. I perform a further robustness check specific to Solar Carve-Out using alternative assumptions about system owners' expectations of SREC value. An owner with rational expectations would perfectly predict future SREC prices and sell SRECs once a year at that year's mean price. However, it is hard for an individual to predict the actual price of SRECs, especially in the initial years after the launch of Solar Carve-Out II. The clearinghouse auction price was intended to support SREC prices, but for a period SRECs consistently traded below it because of oversupply (newly installed capacity quickly surpassed the annual target set by the RPS). The common practice of contracting aggregators or brokers to handle SREC trades makes the actual SREC sale price even more difficult to predict. I therefore consider three ways of predicting SREC prices in addition to actual market prices. Optimistic customers use the ACP—the theoretical upper bound—as the predicted SREC price. Pessimistic customers use the clearinghouse auction price. Neutral customers use the average of the ACP and the clearinghouse auction price, which is the baseline. A standard 10% transaction fee applies to both optimistic and neutral customers. Pessimistic customers are not subject to any other fees, because the transaction fee is already embedded in the clearinghouse auction price.

Table 12 presents pass-through estimates for Solar Carve-Out projects under different SREC price assumptions. Only the Solar Carve-Out sample is shown, and the results are again very close to each other. The highest estimated pass-through rate comes from using the alternative compliance payment as the SREC price, and the lowest from using auction prices. The ordering is as expected, because the ACP is the upper bound of SREC prices and yields the largest subsidy, while the auction price is the lowest assumed path and yields the smallest. The estimates range from 0.723 to 0.782. The estimate using actual market prices, 0.747, is close to the baseline of 0.756. Although I cannot determine owners' true beliefs about the income stream SRECs will actually generate, the results show that the

Solar Carve-Out estimate is robust to the assumed SREC price path.

Table 12: Pass-Through under Alternative SREC Price Paths

Dependent Variable:	Net Price per kW			
Model:	(1) ACP	(2) Auction	(3) Actual price	(4) Midpoint
Variables				
Subsidy per kW	−0.782***	−0.723***	−0.747***	−0.756***
	(0.039)	(0.050)	(0.046)	(0.044)
Installer	✓	✓	✓	✓
Inverter	✓	✓	✓	✓
Manufacturer	✓	✓	✓	✓
City	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491

Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. Subsidies are discounted at 8% with 0.5% annual panel degradation.

Discount rates. Because the subsidy is a discounted sum of future cash flows, the discount rate affects both the dependent variable (y_{it}) and the regressor (Subsidy_{it}). De Groote and Verboven (2019) find that when subsidies are delivered through electricity production, consumers apply high discount rates to future benefits. They estimate an average rate of 15% in Belgium, well above my baseline assumption and above the contemporaneous risk-free euro interest rate, which remained below 1% during their study period. I therefore also use rates of 3%, 5%, and 15%.

Table 13 presents pass-through estimates under varying discount rate assumptions. In Panel A, the estimated pass-through rate declines from 0.805 at a 3% discount rate to 0.682 at 15%. In Panel B, the estimate is essentially constant, between 1.013 and 1.022. The discount rate mechanically changes the size of the regressor and the dependent variable. For SMART the change is purely mechanical. The subsidy is $r_i g_i A(d)$ (Section 3.4), so changing d multiplies every subsidy by the same factor. The coefficient on the subsidy in the gross price, $1-\rho$, is divided by that factor, and $(\hat{\rho}_d - 1)A(d)$ is the same at every discount rate. With $A(8\%)/A(3\%) \approx 0.79$, the baseline estimate implies $\hat{\rho}_{3\%} - 1 \approx 0.017 \times 0.79 \approx 0.013$, which

is the estimate in the table, and the test of $\rho = 1$ gives the same p-value (about 0.80) at every rate by construction. The SMART results therefore carry no information about how buyers discount the subsidy. Only for Solar Carve-Out, where the price path differs across cohorts, does the discount rate change the relative subsidy of different projects, and at every discount rate the Solar Carve-Out estimate is significantly below one (Table 14).

Table 14 reports the corresponding tests of complete pass-through. For SMART, the standard error times $A(d)$ is also the same at every rate, and so is the p-value.

Table 13: Pass-Through at Alternative Discount Rates

	Panel A: Solar Carve-Out				Panel B: SMART			
	Dependent Variable: Net Price per kW				Dependent Variable: Net Price per kW			
	(1) d=3%	(2) d=5%	(3) d=8%	(4) d=15%	(5) d=3%	(6) d=5%	(7) d=8%	(8) d=15%
Variables								
Subsidy per kW	−0.805***	−0.786***	−0.756***	−0.682***	−1.013***	−1.015***	−1.017***	−1.022***
	(0.035)	(0.039)	(0.044)	(0.058)	(0.051)	(0.056)	(0.065)	(0.086)
Installer	✓	✓	✓	✓	✓	✓	✓	✓
Inverter	✓	✓	✓	✓	✓	✓	✓	✓
Manufacturer	✓	✓	✓	✓	✓	✓	✓	✓
City	✓	✓	✓	✓	✓	✓	✓	✓
Utility FE					✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓	✓
Observations	25,491	25,491	25,491	25,491	14,317	14,317	14,317	14,317

Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1. 0.5% annual panel degradation. SREC subsidies use the ACP–auction midpoint price path.

Table 14: Test of $\rho = 1$ by Program and Discount Rate

	d=3%	d=5%	d=8%	d=15%
Panel A: Solar Carve-Out				
Subsidy per kW	−0.805***	−0.786***	−0.756***	−0.682***
	(0.035)	(0.039)	(0.044)	(0.058)
p-value, $\rho = 1$	< 0.001	< 0.001	< 0.001	< 0.001
Panel B: SMART				
Subsidy per kW	−1.013***	−1.015***	−1.017***	−1.022***
	(0.051)	(0.056)	(0.065)	(0.086)
p-value, $\rho = 1$	0.795	0.795	0.795	0.796

Coefficient on subsidy per kW from the baseline specifications. The dependent variable is the net price per kW, and pass-through ρ is minus the coefficient. The p-values test complete pass-through ($\rho = 1$, a coefficient of −1), using installer-clustered standard errors. Stars test the coefficient against zero. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.6 Pooled versus Separate Estimation

Table 15 compares the separate estimates with pooled ones. Columns (1) and (2) come from one stacked regression in which the slope, the controls, and every fixed effect are specific to each program, clustered by installer across programs. Its estimates equal those from separate regressions, and the difference between the programs, reported below the two panels, has a standard error that accounts for installers active in both. Columns (3) to (6) impose one slope. With program-specific fixed effects (columns 3 and 4), the pooled slope is a weighted average of the two program slopes, with weights proportional to the residual variance of the subsidy in each program. Solar Carve-Out receives two-thirds to three-quarters of the weight because its sample is larger. With fixed effects shared across programs (columns 5 and 6), installer, equipment, and town effects are also assumed to be constant from 2014 to 2023.

Table 15: Pooled versus Separate Estimation

	Dependent variable: net price per kW					
	Slope by program		One slope, program FE		One slope, shared FE	
	Baseline (1)	Cross-roof (2)	Baseline (3)	Cross-roof (4)	Baseline (5)	Cross-roof (6)
Panel A: Solar Carve-Out						
Subsidy per kW	−0.756***	−0.863***				
	(0.044)	(0.044)				
Panel B: SMART						
Subsidy per kW	−1.017***	−0.847***				
	(0.065)	(0.072)				
Difference in pass-through (SMART − SCO)	0.260***	−0.016				
	(0.090)	(0.085)				
Standard error if programs independent	(0.078)	(0.084)				
Subsidy per kW, both programs			−0.843***	−0.859***	−0.897***	−0.858***
			(0.031)	(0.037)	(0.038)	(0.037)
Weight on Solar Carve-Out			0.666	0.736		

Entries are coefficients on subsidy per kW, except the difference row, which reports the difference in pass-through. Pass-through is minus the coefficient. Baseline: the specification of column (1) of Table 4. Cross-roof: column (4). Standard errors clustered by installer across programs in parentheses. The weight on Solar Carve-Out is its share of the residual variance of the subsidy after the fixed effects and controls. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Pless and van Benthem (2019) pool host-owned and third-party systems, which share one program, period, subsidy schedule, and set of fixed effects. The two programs here differ in

all four, so I let every coefficient differ by program. Four considerations support this choice. First, pooling imposes equal pass-through, which the baseline rejects ($p = 0.004$). The pooled baseline rate of 0.84 is then an average that describes neither program, and its weights reflect sample sizes rather than any policy question. Second, the paper's question is whether the two designs differ, which requires a slope for each. Third, the subsidies are measured differently. The SMART subsidy is a fixed contract, whereas the Solar Carve-Out subsidy depends on an assumed SREC price path, and pooling would let that assumption affect the SMART estimate. Fourth, the fixed effects must differ by program anyway. SMART needs utility-by-time effects because its rates differ by utility, the programs overlap only in 2018, and installer, town, and equipment effects plausibly changed between the two periods. Sharing them raises the baseline pooled rate from 0.84 to 0.90, because it lets comparisons across programs, in which the subsidy fell by a factor of 3.4 while costs were falling, enter the estimate.

In the preferred specification, the two program estimates are nearly identical, and every version gives about 0.86. Pooling is then harmless, and 0.86 (0.04) is a fair summary of both programs.

B.7 Alternative Identification

The baseline model identifies pass-through from all sources of variation in the subsidy, including trends that could move together with costs. This section reports two designs that rely on discrete, rule-based changes in the subsidy. Both use the commercial operation date as the timing variable and compare projects that differ in the subsidy they face over a short interval.

A regression discontinuity around the January 2017 cut in the SREC factor (from 1.0 to 0.8) gives Solar Carve-Out pass-through of 0.54 to 0.89 when the pre-deadline rush of completions in December 2016 is excluded (0.71 in the preferred specification) and 0.71 to 1.11 when it is included: similar in magnitude to the baseline, but sensitive to specification.

A stacked event study around nine step-downs in SMART rates finds no break in the gross price and pooled pass-through of about 1.27 (standard error 0.15), consistent with near-complete pass-through but imprecise, because the rate is locked at approval while the data record the date the system began operating. A point estimate above one is, as discussed in Section 5.2, more plausibly a sign that the step-downs coincide with falling costs than evidence of over-shifting. Both designs still compare projects a few months apart, so they complement rather than replace the cross-roof estimates.

B.7.1 The January 2017 SREC Factor Cut

On January 8, 2017, the SREC factor for new small systems fell from 1.0 to 0.8 (Section 2). A system that generates the same electricity receives 20% fewer certificates after the cutoff, so the subsidy per kW falls discretely without any change in the installer's costs. Figure 4 plots monthly means by month of operation. The subsidy per kW falls from about $2,110 in December 2016 to about $1,610 from February 2017, and every project operating after the cutoff has a factor of 0.8, with a single exception.

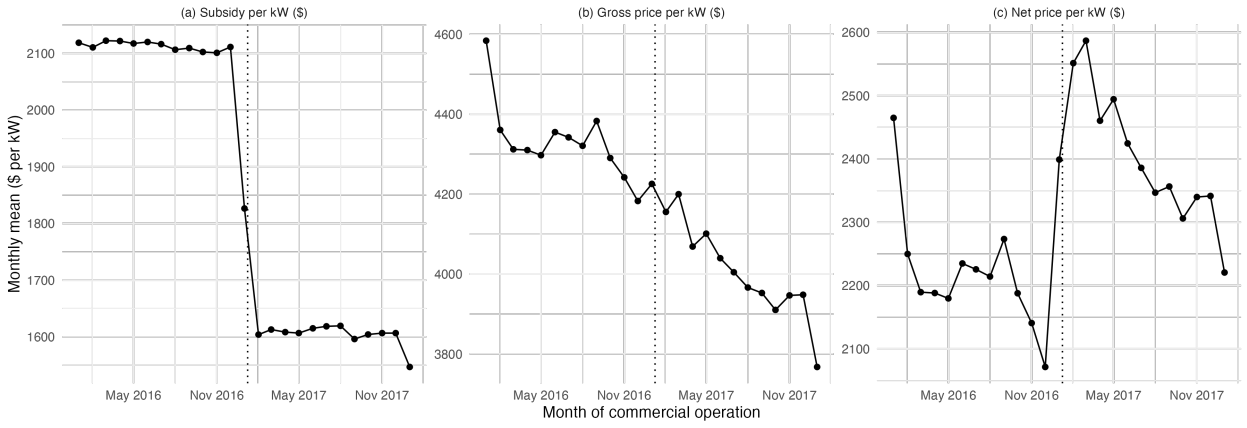


Figure 4: Subsidy, Gross Price, and Net Price by Month of Operation around the SREC Factor Cut

Two features of the raw data complicate the comparison. First, gross prices fall throughout 2016–2017, by roughly $230 per kW per year, so I control for a linear trend on each side of the cutoff. Second, the number of projects operating in December 2016 (879) is well above

that in each of the eleven preceding months (358–647), which suggests a rush to complete before the deadline, and December projects have unusually low gross prices (about $4,180 per kW, compared with $4,240–$4,380 in the preceding months). I therefore report estimates with and without the two months around the cutoff.

Table 16 reports the discontinuity estimates. The subsidy falls by $465–$500 per kW in the baseline measure and $420–$435 in the factor-only measure, which holds the SREC price path fixed and isolates the change caused by the factor. Excluding December 2016 and January 2017, pass-through estimates lie between 0.54 and 0.89, with a preferred specification (a window of 180 days and the factor-only subsidy) of 0.71 (standard error 0.12). Including those months, the estimates are higher and more variable (0.71 to 1.11), which is consistent with the low December prices widening the apparent jump in net prices. The estimates are noisy at the narrowest bandwidth (a standard error of 0.21–0.25 when the rush is excluded). Overall, this design points to incomplete pass-through under Solar Carve-Out of similar size to the baseline estimate of 0.76, although the range across specifications is wide, and I do not treat any single estimate as more than suggestive.

B.7.2 Step-Downs in SMART Rates

The same logic applies to SMART: when a capacity block fills, the base rate for the utility falls by a known amount, and the subsidy per kW falls discretely. I identify step-downs from the monthly modal SMART rate: a step is a month in which the modal rate falls by more than $0.004 per kWh. Reading the rate this way requires many projects in each month, so I use the two utilities that account for 90% of SMART projects, Eversource East (7,400 projects, 51 months with at least 15 projects) and National Grid (5,747 projects, 46 such months). Unitil has 88 projects and never reaches 15 in a month. Eversource West has 1,082 projects, and its modal rate is noisy in the months with few installations. Adding it gives twelve steps and a pass-through estimate of 1.26 (0.15), almost identical to the estimate below. I keep nine steps that are at least four months from any other step. Each step lowers

Table 16: Discontinuity at the January 2017 SREC Factor Cut

	All projects			Excluding Dec. 2016–Jan. 2017		
Bandwidth (days)	±90	±180	±365	±90	±180	±365
	(1)	(2)	(3)	(4)	(5)	(6)
Subsidy per kW, baseline measure (first stage)	−466***	−478***	−478***	−495***	−497***	−489***
	(12)	(9)	(8)	(34)	(13)	(10)
Subsidy per kW, factor-only measure (first stage)	−434***	−430***	−424***	−425***	−427***	−422***
	(11)	(9)	(8)	(34)	(13)	(11)
Gross price per kW	16	−65**	−139***	−117	−192***	−227***
	(46)	(33)	(37)	(104)	(50)	(46)
Net price per kW	482***	413***	339***	378***	305***	263***
	(48)	(34)	(41)	(114)	(51)	(49)
Pass-through $\hat{\rho}$, baseline measure	1.034***	0.864***	0.708***	0.763***	0.613***	0.537***
	(0.098)	(0.068)	(0.079)	(0.213)	(0.100)	(0.096)
Pass-through $\hat{\rho}$, factor-only measure	1.110***	0.960***	0.799***	0.888***	0.713***	0.622***
	(0.105)	(0.076)	(0.089)	(0.246)	(0.117)	(0.110)
Observations	3,206	5,995	11,795	1,722	4,527	10,328

Solar Carve-Out projects. Each column is a set of regressions on projects within the stated window around January 8, 2017, using commercial operation date as the running variable, with a linear trend on each side of the cutoff and installer, inverter, manufacturer, and city fixed effects. Entries for the first four rows are the estimated jump at the cutoff in dollars per kW. Pass-through is estimated by two-stage least squares with the post-cutoff indicator as the instrument for the subsidy. The factor-only measure holds the present-value factor of the SREC price path at its 2016 value, so that the subsidy varies only through the SREC factor and generation. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

the rate by \$0.011–\$0.020 per kWh. I stack the nine events and estimate the change in subsidy, gross price, and net price in the four months before and three months after each one, relative to the month before the step.

The data record the date a project began operating, while the SMART rate is fixed when the application is approved, which is usually months earlier. The subsidy therefore does not jump at the step but declines gradually across the window (Figure 5, panel a): it is about \$51 per kW above its month −1 level four months before the step and \$88 below it three months after. If pass-through is complete, the gross price should not respond, and the net price should rise by the amount the subsidy falls. Panel (b) shows that the gross price does not break at the step. Its event-time coefficients lie between −\$40 and +\$26 per kW, with confidence intervals of roughly ±\$40, and panel (c) shows that the net price rises as the subsidy falls.

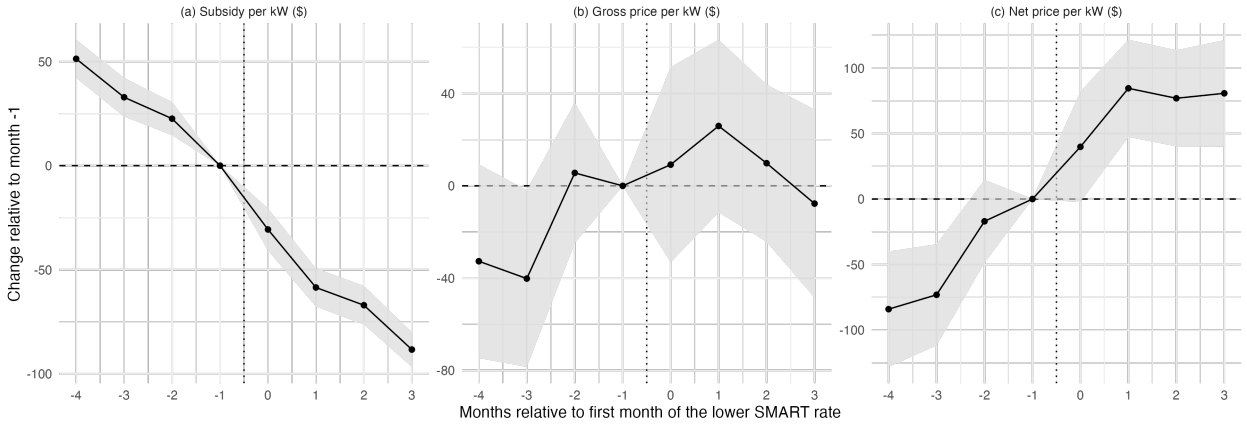


Figure 5: Event Study around SMART Rate Step-Downs

Table 17 pools the post-step periods. The subsidy is \$85 per kW lower after a step, the gross price is \$23 higher (a standard error of \$13), and the net price is \$108 higher, which implies pass-through of 1.27 (standard error 0.15) when the post indicator instruments the subsidy. Adding a linear trend in event time yields an estimate of 1.65 with a standard error of 1.43, so the design cannot identify pass-through once the gradual decline in the subsidy is absorbed by the trend. Read together, the event study is consistent with pass-through

near or above one at SMART step-downs, in line with the baseline estimate of 1.02, but it is imprecise, it rests on nine events in two utilities, and it cannot rule out that some step-downs coincide with updates to the value of energy.

Table 17: Pooled Event Study around SMART Rate Step-Downs

	No trend	Linear event-time trend
	(1)	(2)
Subsidy per kW, post vs. pre ($)	−85.2***	−18.2***
	(2.4)	(5.8)
Gross price per kW, post vs. pre ($)	23.2*	11.8
	(12.9)	(25.7)
Net price per kW, post vs. pre ($)	108.4***	30.0
	(12.8)	(26.1)
Pass-through, IV	1.272***	1.645
	(0.153)	(1.435)
Observations	10,851	10,851

SMART projects. Stacked sample of nine step-downs in the modal SMART rate (two National Grid, seven Eversource East), with projects from four months before to three months after the first month of the lower rate. Post is the indicator for event time $k \geq 0$, and column (2) adds a linear trend in event time. Regressions include event, installer, inverter, manufacturer, and city fixed effects. Pass-through is estimated by two-stage least squares with the post indicator as the instrument for the subsidy. Standard errors clustered by installer in parentheses. Discount rate 8%. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.8 Heterogeneity by Income and Installer Concentration

I examine whether SMART pass-through differs across ZIP codes by median household income and by local installer concentration. Income is the ZIP-level median household income from the American Community Survey (2016–2020 five-year estimates). Concentration is the Herfindahl–Hirschman index (HHI) of installers' shares of SMART projects in the ZIP code. I divide ZIP codes into terciles on each measure. For each measure, I estimate a single regression that interacts the subsidy with tercile indicators and keeps the fixed effects common across groups, and then test whether the three coefficients are equal. Table 18 reports the results.

Table 18: Heterogeneity in SMART Pass-Through

	Dependent variable: net price per kW							
	ZIP median income tercile				Installer HHI tercile			
Discount rate	3%	5%	8%	15%	3%	5%	8%	15%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Subsidy per kW × tercile 1	−1.074***	−1.081***	−1.094***	−1.125***	−0.968***	−0.965***	−0.959***	−0.946***
	(0.057)	(0.062)	(0.072)	(0.096)	(0.062)	(0.068)	(0.079)	(0.105)
Subsidy per kW × tercile 2	−1.024***	−1.026***	−1.030***	−1.040***	−1.032***	−1.035***	−1.040***	−1.054***
	(0.055)	(0.061)	(0.070)	(0.094)	(0.053)	(0.059)	(0.068)	(0.090)
Subsidy per kW × tercile 3	−0.929***	−0.922***	−0.910***	−0.880***	−1.045***	−1.050***	−1.057***	−1.077***
	(0.054)	(0.060)	(0.069)	(0.092)	(0.051)	(0.057)	(0.065)	(0.087)
p-value, equal pass-through	0.011	0.011	0.011	0.011	0.108	0.108	0.108	0.108
Observations	14,195	14,195	14,195	14,195	14,195	14,195	14,195	14,195

SMART projects. Each column is one regression, with the subsidy per kW interacted with tercile indicators. Tercile 1 is the lowest-income tercile in columns (1)–(4) and the least concentrated tercile in columns (5)–(8). Entries are coefficients, so pass-through is minus each entry. All regressions include utility, installer, inverter, manufacturer, city, and year fixed effects. Standard errors clustered by installer in parentheses. The p-value is from a joint Wald test that the three interaction coefficients are equal. It is the same at every discount rate because the discount rate rescales every SMART subsidy by the same factor (Appendix B.5). The sample has 14,195 projects, compared with 14,317 in the full SMART sample. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Pass-through falls with income. At the baseline discount rate, the estimates are 1.09 in the lowest tercile, 1.03 in the middle tercile, and 0.91 in the highest tercile, and the hypothesis that they are equal is rejected ($p = 0.011$). The pattern is mechanically the same at every discount rate (Appendix B.5). By concentration, the estimates rise slightly from 0.96 in the least concentrated tercile to 1.06 in the most concentrated, the opposite of what market power would predict, and the equality test does not reject ($p = 0.108$).

Two cautions apply. First, the pattern is not robust to specification. If I estimate the model separately within each tercile, allowing every fixed effect to differ by group, the income gradient disappears: the estimates are 0.97, 1.11, and 0.97 from lowest to highest income tercile, and 1.02, 1.10, and 0.94 across concentration terciles, with no monotone pattern (Figures 6 and 7). The pooled interaction model imposes common installer, city, and equipment effects, which is the appropriate comparison when the question is whether the subsidy coefficient differs given the same cost structure, but the split-sample results show that the gradient depends on this choice, and I do not put weight on it. Second, income and concentration are measured at the ZIP level from SMART records only. Income is an ACS estimate for the ZIP, not the buyer's own income, and the HHI is computed from the same projects used in the regression, so ZIPs with few projects have mechanically high concentration. Terciles are equal-sized groups of projects and not of ZIP codes.

Taken together, the results are at most suggestive that pass-through is lower for buyers in higher-income areas, and they give no evidence that pass-through is lower where installers are more concentrated. The second finding bears on the interpretation of the main results: if pass-through under SMART reflected market power, one would expect it to vary with local competition and, since demand is log-concave (Section 6.2), to be lower where competition is weaker. The absence of this gradient is consistent with the view that market power is not the main determinant of pass-through in this market, but the test has limited power, since the HHI measure varies within a range of low concentration statewide (Appendix C.3).

Figures 6 and 7 report SMART pass-through when the model is estimated separately within income and concentration terciles, so that every fixed effect can differ by group.

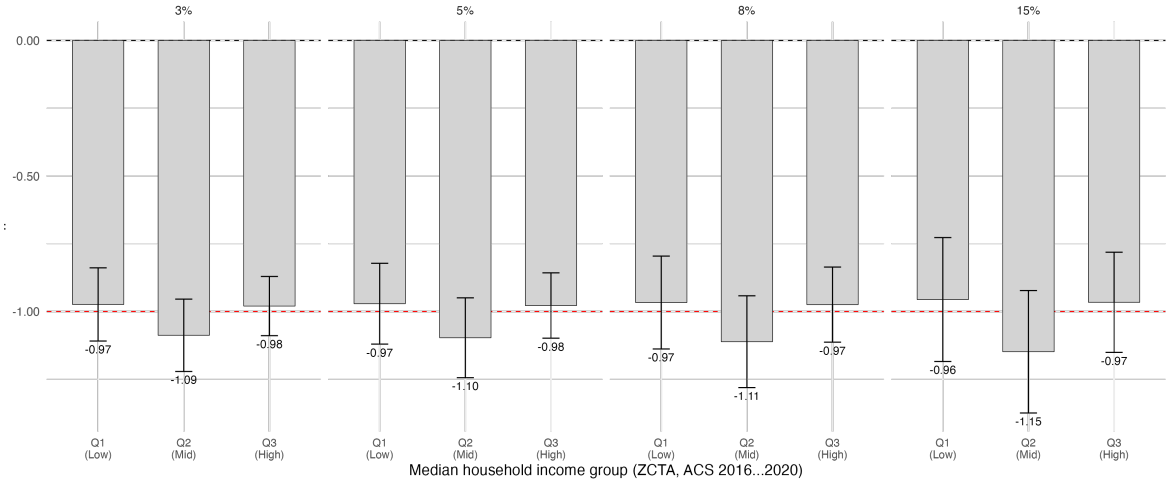


Figure 6: SMART Pass-Through by ZIP Income Tercile, Separate Regressions

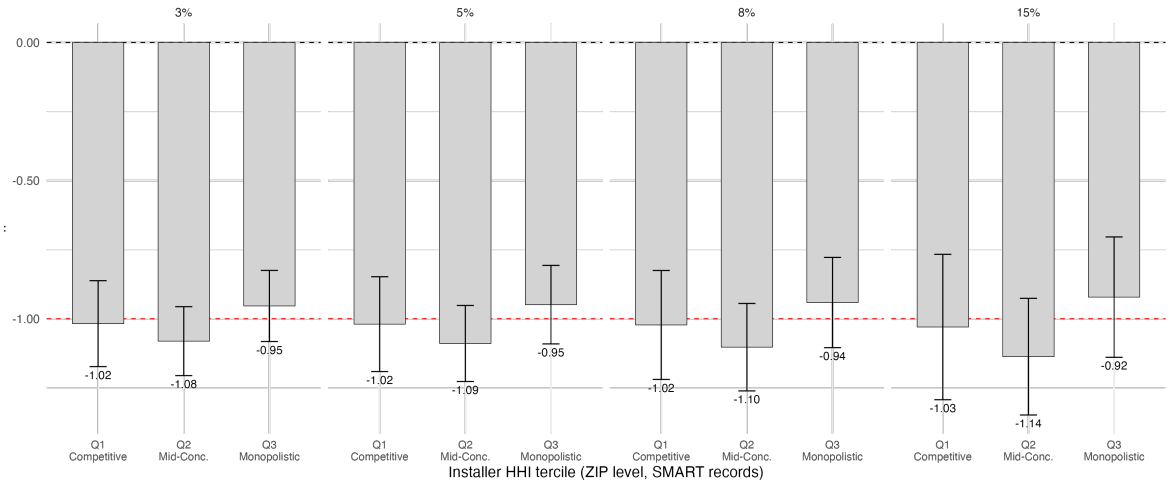


Figure 7: SMART Pass-Through by Installer Concentration Tercile, Separate Regressions

B.9 Demand Estimation: Alternative Specifications

Table 19 reports the quadratic demand estimates of Section 6.2 under alternative specifications. With county-by-year fixed effects and only the subsidy instruments, the log-quadratic SMART estimates are almost unchanged: the coefficient on the squared price is −0.425 (0.103) and the curvature is 2.13 (1.12). Under SMART, the price coefficients and the curvature keep their sign in every specification. Under Solar Carve-Out, the estimates are imprecise and change sign, so demand is not identified.

Table 19: Demand Estimates under Alternative Specifications

	Baseline	Utility FE added	County × year FE, subsidy IV	Population-weighted	ZIP-years with 3+ installations	Clustered by county
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Solar Carve-Out						
Price elasticity at the mean	0.27	0.30	−2.51	2.24	−4.93**	0.27
	(1.21)	(1.29)	(1.61)	(1.40)	(2.46)	(1.12)
Net price squared (β_2)	−0.124	−0.216	−0.202	−1.001	5.026**	−0.124
	(0.716)	(0.765)	(0.909)	(0.857)	(2.132)	(0.768)
Curvature $1/\varepsilon_{ms}$ at the mean	14.86	19.58	1.25	2.59***	−0.42	14.86
	(56.01)	(97.34)	(1.43)	(0.78)	(0.91)	(58.00)
ZIP-years	2,171	2,171	2,171	2,171	1,809	2,171
Panel B: SMART						
Price elasticity at the mean	−1.57***	−1.92**	−1.62***	−1.21***	−1.14**	−1.57***
	(0.42)	(0.77)	(0.43)	(0.37)	(0.47)	(0.51)
Net price squared (β_2)	−0.439***	−0.417***	−0.251***	−0.216***	−0.570***	−0.439***
	(0.082)	(0.092)	(0.082)	(0.062)	(0.116)	(0.122)
Curvature $1/\varepsilon_{ms}$ at the mean	5.38*	3.80	3.35*	4.64	9.78	5.38
	(2.77)	(2.65)	(1.73)	(2.97)	(8.39)	(3.55)
ZIP-years	1,715	1,715	1,715	1,715	1,237	1,715

Two-stage least squares estimates of equation (2). Elasticity and curvature at the mean net price and installation rate of each sample. Baseline: county and year fixed effects, with the subsidy per watt, the county building-equipment contractor wage, and their squares as instruments. Column (3) uses only the subsidy instruments, because the wage varies at the county-year level. Standard errors clustered by ZIP code (by county in column 6) in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

B.10 Demand Estimation: Flexible Functional Form

The quadratic in Section 6.2 estimates curvature from the data but restricts how curvature varies with price. To check whether this restriction matters, I replace the quadratic with a natural cubic spline in the net price, with three to five degrees of freedom and interior knots at quantiles of the price distribution. A spline enters the endogenous price through several nonlinear terms, and two-stage least squares with polynomials of the instruments identifies them only weakly: with nine instruments, the curvature at the mean ranges from −87 to 1.9 across three to five degrees of freedom, with standard errors of up to 389. I therefore use a control function (Newey, Powell and Vella, 1999). The first stage regresses the price on the instruments, controls, and fixed effects of equation (2), and the demand equation adds the first-stage residual to the spline. Standard errors come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. Under Solar Carve-Out, the curvature estimates have standard errors in the thousands under every form, consistent with demand not being identified (Section 6.2), so Table 20 reports SMART only.

Three findings stand out. First, curvature is positive in every specification, between 0.73 and 1.51 at the mean price, and 97 to 100 percent of bootstrap draws imply log-concave demand. The conclusion that demand under SMART is log-concave therefore does not depend on the quadratic. Second, with the estimator held fixed, the quadratic and the splines give similar curvature (1.31 for the quadratic and 0.73 to 1.51 for the splines), so the quadratic is flexible enough near the mean. Third, the baseline curvature of 5.38 is larger than under any flexible form. Figure 8 shows why. The baseline quadratic turns upward at \$2.87 per watt, not far below the mean of \$3.35, and the curvature of a quadratic rises steeply as its slope approaches zero. From about \$3.80 per watt upward, all three curves in the figure lie between about one and two.

Smaller curvature loosens the bound. At the cross-roof pass-through estimate, the implied markup is at most 4.1 to 7.0 percent of price under the control-function estimates, compared with 2.1 percent for the baseline quadratic and 3.9 percent for the log-quadratic

form. Markups remain a few percent of price, but the baseline quadratic gives the tightest bound.

Table 20: SMART Demand Curvature under Flexible Functional Forms

	Two-stage least squares		Control function			
	Quadratic (baseline)	Log-quadratic	Quadratic	Natural spline		
				3 df	4 df	5 df
	(1)	(2)	(3)	(4)	(5)	(6)
Elasticity at the mean	−1.57***	−2.16***	−2.54***	−3.01***	−2.94***	−3.52***
	(0.42)	(0.42)	(0.36)	(0.41)	(0.39)	(0.76)
Curvature $1/\varepsilon_{ms}$ at the mean	5.38*	2.14**	1.31***	1.32***	1.51***	0.73
	(2.77)	(1.02)	(0.17)	(0.14)	(0.56)	(0.98)
Share of bootstrap draws log-concave			1.00	1.00	1.00	0.97
Markup bound (% of price)	2.1	3.9	5.5	4.6	4.1	7.0

SMART ZIP-years (1,715). Elasticity and curvature at the mean net price and installation rate. Columns (1) and (2): two-stage least squares from Table 5, with standard errors clustered by ZIP code. Columns (3)–(6): control function, in which the demand equation includes the first-stage residual of the price. Natural splines have interior knots at quantiles of the net price. Standard errors in columns (3)–(6) and the share of log-concave draws come from 499 bootstrap draws that resample ZIP codes and re-estimate both stages. The markup bound is $(1/\rho_B - 1)/(1/\varepsilon_{ms})$ divided by the absolute elasticity, at $\rho_B = 0.847$ (Table 4, column 4). All specifications include the ZIP controls and county and year fixed effects of equation (2). Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

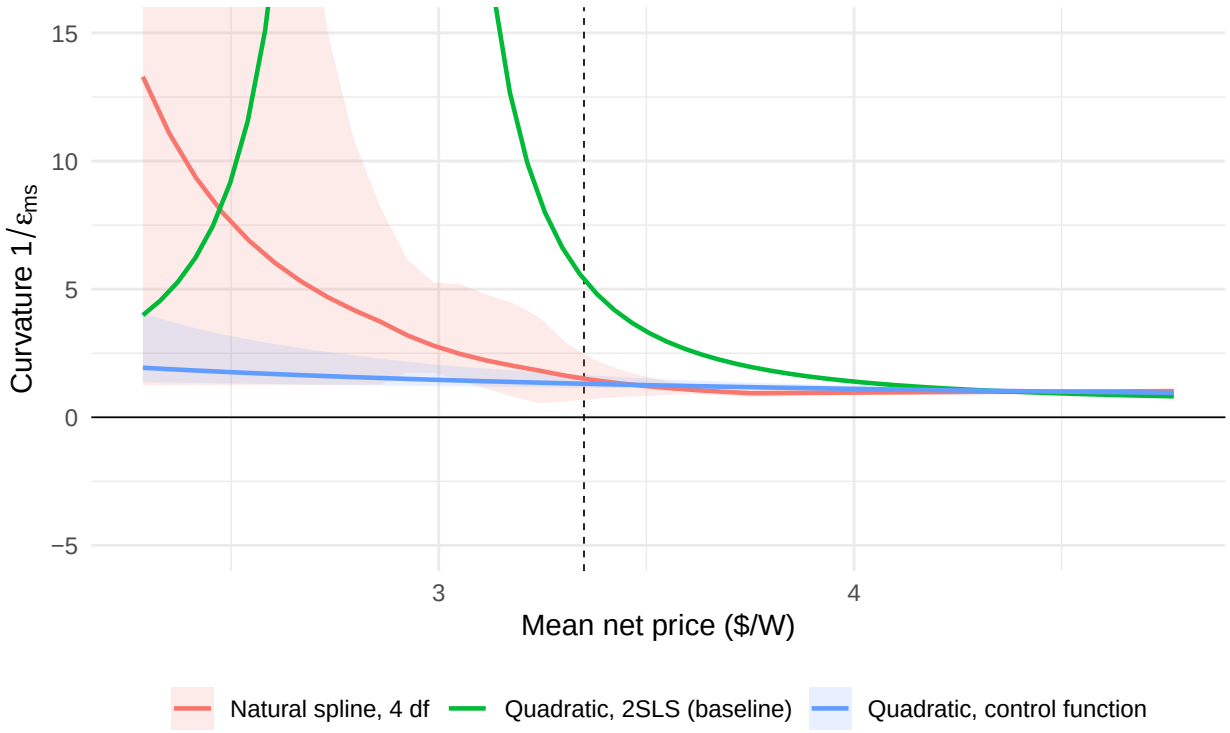


Figure 8: Curvature of SMART Demand across Prices. Curvature $1/\varepsilon_{ms}$ evaluated along each fitted demand curve between the 5th and 95th percentiles of the net price. Shaded areas are 90% bootstrap intervals for the control-function estimates. The dashed line marks the mean net price.

C Other Explanations for the Baseline Gap

The baseline estimates differ between the programs: 0.76 for Solar Carve-Out and about one for SMART. Section 5.2 shows that most of this gap disappears once the timing of the subsidy is absorbed, so the main explanation is how each subsidy moved over time relative to installation costs. This section examines three further candidate explanations for the baseline gap. None of the three can be isolated cleanly, because the programs differ in size, certainty, period, and eligible population at the same time, so the evidence here is descriptive.

C.1 Size

The first explanation is that SMART subsidies are small enough to be absorbed without a supply response, whereas Solar Carve-Out subsidies, worth about 45% of the gross price, strain installers' capacity. The mean subsidy is $1,923 per kW under Solar Carve-Out and $566 per kW under SMART, and Section 3 shows that the supports of the two distributions barely overlap. A direct comparison at the same subsidy level is therefore not possible. What I can do is ask whether pass-through falls with the size of the subsidy within each program. Table 21 reports pass-through by quartile of subsidy per kW. Under Solar Carve-Out the estimate is flat, between 0.77 and 0.78 in every quartile, so within that program larger subsidies are not passed through less. Under SMART the estimate falls from 1.55 in the lowest quartile to 1.09 in the highest, which is the direction the size explanation predicts. However, the lowest SMART quartile has a median subsidy of $127 per kW and includes projects with no subsidy at all, so the estimate is imprecise (a standard error of 0.23), and the gradient may also reflect falling costs as the subsidy shrinks across blocks. Estimates above one in the two lowest quartiles are hard to reconcile with log-concave demand (Section 5.2), which favors the second reading. Taken together, the within-program evidence is mixed: it does not support the size explanation under Solar Carve-Out and is consistent with it, but confounded, under SMART. The workload test in Section 6.4 points the same way for Solar

Carve-Out: if large subsidies strained installers' capacity, pass-through would be lower when installers are busy, and it is not.

Table 21: Pass-through by Quartile of Subsidy per kW

	Q1	Q2	Q3	Q4
Panel A: Solar Carve-Out				
Median subsidy ($/kW)	1,528	1,698	2,151	2,305
Subsidy per kW	−0.779***	−0.776***	−0.780***	−0.770***
	(0.073)	(0.068)	(0.062)	(0.059)
Panel B: SMART				
Median subsidy ($/kW)	127	458	669	947
Subsidy per kW	−1.552***	−1.317***	−1.189***	−1.091***
	(0.228)	(0.131)	(0.091)	(0.076)

Coefficient on subsidy per kW in each quartile, from a regression of the net price per kW in which subsidy per kW is interacted with quartile indicators of the same variable, with the full set of fixed effects. Pass-through is minus the coefficient. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

C.2 Predictability

The second explanation is that fixed SMART rates reduce uncertainty. Buyers who cannot forecast SREC revenue may value it less than its expected present value, and installers cannot raise prices as far. Three observations bear on this. First, the ACP capped the SREC price and the auction price was meant to support it, so the range of likely prices was narrower than in an unregulated market. Market prices nonetheless fell below the auction price for several years (Figure 1), so the auction price did not act as a floor in practice. Second, the estimate is not sensitive to how one forecasts prices: it ranges from 0.723 (auction price) to 0.782 (ACP) across the three assumptions and is 0.747 with actual market prices (Table 12). If uncertainty about the price path were the main driver, the estimate should be more sensitive to the assumed path. Third, pass-through under Solar Carve-Out falls with the discount rate (0.805 at 3% versus 0.682 at 15%), whereas the SMART estimate cannot respond to it by construction (Appendix B.5). The Solar Carve-Out pattern is consistent with buyers applying high discount rates to uncertain, back-loaded certificate revenue, but

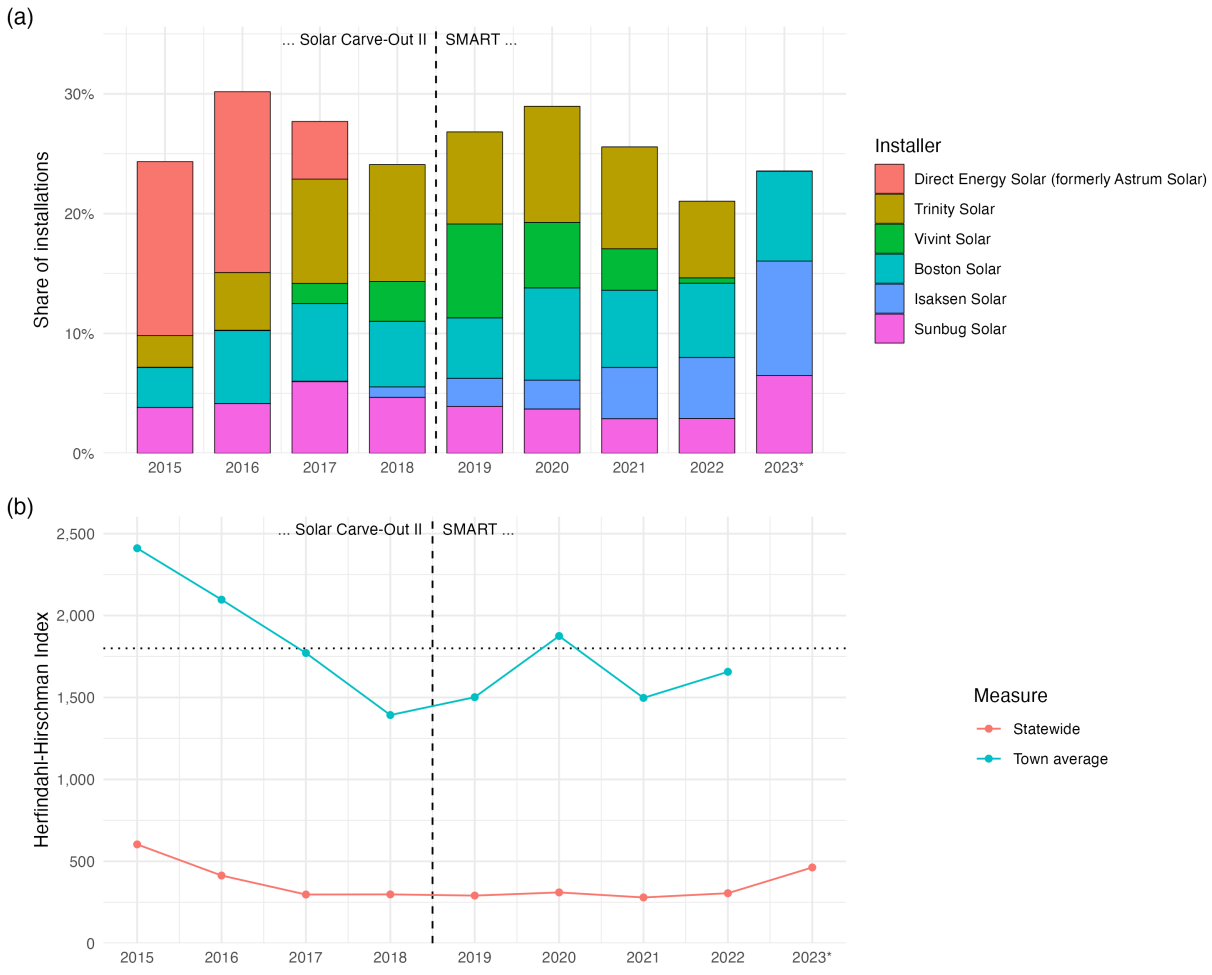


Figure 9: Installer Market Share and Concentration

it is also partly the mechanical effect of a larger regressor.

C.3 Market Competition

The third explanation concerns competition. Figure 9 shows the concentration of installers over the sample. Panel (a) shows the market share of the four largest installers in each program, and panel (b) shows the Herfindahl–Hirschman index (HHI) at the statewide and town levels.

At the state level, the market is unconcentrated throughout. The statewide HHI is 603 in 2015 and falls to about 300 by 2017, where it stays between 280 and 310 through 2022.

The top four installers' combined share falls from about 40% in 2015 to 23–27% from 2017 on. The number of active installers per year is 184–216 in 2015–2018 and 128–162 in 2019–2022. The figure starts in 2015 because Solar Carve-Out II opened in April 2014 and, in that partial year, only 28 towns have at least 20 installations, so the 2014 town average (above 6,000) is not comparable with later years. Town-level concentration, which is more relevant for a household choosing among installers who quote locally, is higher: the project-weighted town HHI falls from about 2,400 in 2015 to about 1,400 in 2018 and is 1,500–1,900 in the SMART years with no downward trend within the period.

These patterns do not support the idea that the market became less competitive during SMART, nor do they show that competition intensified in a way that could explain the jump in pass-through. Most of the fall in concentration happened before SMART started, between 2015 and 2017, when Solar Carve-Out pass-through was already about 0.76. Within each program, pass-through rises only slightly with local concentration, and the change is not significant at the 5% level (Section 6.4 and Appendix B.8). Together with low statewide concentration, this suggests that market power is not the main reason for the gap between the two programs.

D Price Discrimination Across Roofs

Installers quote each project individually, and a quote typically includes a projection of the roof's output, so the installer knows the subsidy a buyer will receive. Section 6.1 shows that the cross-roof estimates measure the pass-through of a roof-specific subsidy, which is below one only if the gross price rises with the roof's subsidy. This appendix discusses such pricing further.

A production-based subsidy makes this kind of pricing easier than an upfront rebate does. A flat rebate per kW is the same for every roof, whereas a subsidy per kWh varies with a characteristic the installer observes and projects in the quote. Under SMART the rate is also known with certainty at the time of sale. If certainty made pricing by roof easier, cross-roof pass-through would be lower under SMART than under Solar Carve-Out. The two estimates are nearly identical (0.85 and 0.86), which gives no indication that it did.

D.1 Pricing on the Value of Electricity

Pricing by roof need not respond to the subsidy alone. For third-party-owned systems, Bielen et al. (2019) estimate separately how the value of bill savings and of state and local incentives is shared among installers, financiers, and customers, and find that both are passed through to system prices and customer payments. Net metering credits a system's generation at retail rates, so a roof's bill savings B_i are also proportional to its generation. Within the cells that identify the cross-roof estimates, the SMART rate is the same for all buyers, so both the subsidy and the bill savings vary with generation, and $B_i \approx \kappa S_i$ for a ratio κ that is common within the cell. If installers set the gross price to capture a share ϕ of the roof's total value $S_i + B_i$, the estimated pass-through satisfies

$$1 - \hat{\rho}_B = \phi\,(1 + \kappa).$$

Under SMART the incentive rate averages about \$0.07/kWh, while net-metered generation is credited at retail rates several times higher, and bill savings continue after the ten-year SMART term, so κ is large and a small ϕ would produce the whole shortfall. Under Solar Carve-Out the subsidy is about 3.4 times larger, so κ is much smaller. If installers captured the same share of total value in both programs, cross-roof pass-through would be lower under SMART. That it is not suggests that the shortfall is not driven only by pricing on the total value of the roof, but the two programs also differ in period and subsidy uncertainty, so this comparison is not decisive.

D.2 Welfare Effects of Pricing by Roof

If installers do price by roof, a natural question is whether buyers as a group gain or lose relative to a uniform price. Aguirre, Cowan and Vickers (2010) and Miklós-Thal and Shaffer (2021) show that the answer depends on pass-through and on the curvature of demand in each market. Treat roofs with high generation as the strong market, where the discriminatory price is higher, and roofs with low generation as the weak market. Relative to uniform pricing, discrimination raises output if pass-through is higher in the weak market ($\rho_W > \rho_S$, averaged over the relevant range) and inverse demand is convex in both markets. It lowers output if $\rho_W < \rho_S$ and inverse demand is concave in both markets. The same conditions, with pass-through weighted by the markup, determine the welfare effect. Miklós-Thal and Shaffer (2021) show that the ranking of pass-through rates alone is not sufficient, so the sign of curvature is needed. Demand under SMART is concave (Table 5). If pass-through were lower on roofs with low generation, pricing by roof would therefore reduce adoption relative to a uniform price per kW. The first test below finds no such difference, so the data do not sign this effect. Miklós-Thal and Shaffer (2021) also note that formulas based on pass-through hold exactly only for small changes unless demand is linear, so the division of surplus in Section 6.3, which is evaluated at the mean, is an approximation for a subsidy of this size.

D.3 Tests

Two tests separate these channels. Table 22 reports the first and Table 23 the second.

The first test estimates cross-roof pass-through by tercile of modeled capacity factor, using the specification in column (4) of Table 4, with terciles defined within each program. Roofs in the low tercile are the weak market and roofs in the high tercile the strong market. In one regression per program, with a separate slope on the subsidy and a separate intercept for each tercile and common fixed effects, pass-through is nearly the same in all three terciles: 0.98, 0.95, and 0.97 under Solar Carve-Out and 0.87, 0.83, and 0.87 under SMART. Equality is not rejected (p = 0.66 and p = 0.46), and the difference between the high and the low tercile is −0.01 (0.04) in both programs. Estimating column (4) separately on each tercile's projects gives point estimates that fall from the low to the high tercile, from 0.95 to 0.81 under Solar Carve-Out and from 0.96 to 0.79 under SMART, but the standard errors are 0.07 to 0.20 and the differences are not significant. The estimates within terciles are higher than those in column (4) because the tercile intercepts remove the variation in generation between terciles. Measurement error in generation, which pulls the estimate toward one (Section 3.3), has a larger effect on the variation that remains. The data therefore do not show that pass-through differs between the weak and the strong market.

The second test uses SMART projects whose incentive rate is near zero, because the value of energy had nearly caught up with the base rate. There are 980 projects with a rate of at most \$0.01 per kWh, almost all in National Grid territory between 2020 and 2023. In this group the subsidy rises by only about \$2 per kW for each percentage point of capacity factor, compared with about \$43 among other SMART projects, while bill savings still rise with generation. If installers price on the subsidy alone, the gross price in this group should not rise with the capacity factor. If they capture a common share of the roof's total value, the slope should be close to the slope among other SMART projects. Table 23 regresses the gross price per kW on the capacity factor with utility-by-year-month, installer, and city fixed effects and controls for system size and the DC-to-AC ratio. In the near-zero group the slope

is $7 per kW per percentage point (standard error 29) at the $0.01 cutoff and ranges from $7 to $66 across cutoffs, and none of these estimates is significant. Among other SMART projects, the slope is $1.5 to $3 (standard errors of about 4). The test therefore lacks power. Pricing on total value would imply a slope of a few dollars per kW per percentage point, well inside the confidence intervals, so the test cannot distinguish pricing on the value of electricity from pricing on the subsidy.

Table 22: Pass-Through by Tercile of Modeled Capacity Factor

	Dependent variable: net price per kW					
	One regression			Separate regressions		
	Low CF (1)	Middle CF (2)	High CF (3)	Low CF (4)	Middle CF (5)	High CF (6)
Panel A: Solar Carve-Out						
Subsidy per kW	−0.983***	−0.952***	−0.972***	−0.947***	−0.806***	−0.809***
	(0.069)	(0.059)	(0.058)	(0.075)	(0.162)	(0.198)
p-value, equal across terciles		0.658				
Panel B: SMART						
Subsidy per kW	−0.873***	−0.825***	−0.867***	−0.963***	−0.830***	−0.787***
	(0.091)	(0.095)	(0.086)	(0.140)	(0.147)	(0.160)
p-value, equal across terciles		0.459				

Coefficient on subsidy per kW by tercile of modeled capacity factor (CF), defined within each program, in the specification of column (4) of Table 4. Pass-through is minus the coefficient. Columns (1)–(3) come from one regression that interacts the subsidy with tercile indicators and includes tercile intercepts and common fixed effects, and the p-value is from a Wald test that the three slopes are equal. Columns (4)–(6) estimate column (4) of Table 4 separately on each tercile's projects. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table 23: Gross Price and Modeled Generation by SMART Incentive Rate

	Dependent variable: gross price per kW			
Incentive rate cutoff ($/kWh)	= 0	≤ 0.005	≤ 0.01	≤ 0.02
	(1)	(2)	(3)	(4)
Capacity factor (pp), near-zero rate	66.3	48.7	7.0	8.2
	(90.0)	(39.3)	(29.0)	(19.1)
Capacity factor (pp), positive rate	2.1	1.5	1.6	2.9
	(3.9)	(3.9)	(4.2)	(4.5)
Subsidy per kW on capacity factor, near-zero rate	0.0	0.7	2.1	4.2
Projects with near-zero rate	186	616	980	1,886

SMART projects. Coefficient on the modeled capacity factor (percentage points) in a regression of the gross price per kW on projects with an incentive rate at or below the cutoff (near-zero rate) or above it (positive rate), with utility-by-year-month, installer, and city fixed effects, log system size, and DC-to-AC ratio bins. The subsidy row reports the same regression with subsidy per kW as the outcome. Discount rate 8%. Standard errors clustered by installer in parentheses. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.